

Advanced Statistical Learning for Complex Data - Prof. Francesca Ieva
**Network analysis of Visium HD spatial transcriptomics:
topology, niches, and gene co-expression at scale**

Project Report

Matteo D'Oria
Data Analytics & Decision Science PhD candidate
Politecnico di Milano · Human Technopole Foundation

Academic Year 2025/26

Abstract

I analyse a high-resolution spatial transcriptomics cohort of ~ 8 million individually segmented cells from 56 colorectal-cancer patients (112 samples), annotated with eight tumour-microenvironment cell types: B cells, Endothelial cells, Fibroblasts, Myeloid cells, Normal epithelium, SMC, T cells, and Tumor cells. The data are intrinsically network-shaped — every cell has spatial neighbours that condition its biology — and I treat the spatial k -nearest-neighbour graph as the integrating object across a five-Part analysis pipeline.

Part 1 characterises the topology of the spatial graph across 110 non-anomaly samples. Local statistics are remarkably consistent (median mean degree 8.24), but the global fraction of cells in the largest connected component is bimodal, separating compact and fragmented tissue architectures. Topology correlates with cell-type composition: SMC drives sparsity ($\rho = -0.58$ vs. mean degree), and per-cell betweenness identifies a tissue skeleton with no cell-type signature.

Part 2 aligns per-sample Leiden communities into 20 stable niches across the cohort, covering 77% of cohort cells. Niches partition into four recurrence bands: 4 universal, 11 common, 4 selective, and one rare (a pure-SMC niche restricted to 11 patients). The niches map asymmetrically onto the eight cell types: Tumor cells, B cells, and Fibroblasts each dominate four niches; T cells dominate none. The only niche where T cells appear as a substantial secondary minority is Niche 17 (48% B cells, 29% T cells), a composition consistent with tertiary lymphoid structures.

Part 3 runs a five-rung ablation comparing per-cell baselines (logistic regression, multi-modal MLP) to graph-aware architectures (GraphSAGE, GAT) on the cell-type classification task. A 2-layer GraphSAGE on highly variable gene features alone achieves the best validation macro- F_1 (0.6350) and a final test macro- F_1 of 0.6089, a +6.5pp gain over the strongest per-cell baseline. Adding the H&E embedding or graph attention on top of GraphSAGE yields no further improvement: the spatial graph subsumes the architectural signal that the H&E modality contributes in isolation.

Part 4 fits per-class Gaussian Graphical Models on 378 broadly variable common genes. At the near single-cell resolution of Visium HD, mRNA diffusion between adjacent cells means that per-cell expression vectors carry a non-trivial component of neighbourhood signal, and the resulting networks recover *microenvironmental gene programmes* rather than purely cell-intrinsic regulation. Five of the eight classes retain canonical cell-type markers among their top hubs alongside neighbourhood-derived genes; three (B cells, Endothelial cells, T cells) are dominated by neighbourhood signal. Across the eight networks, hub Jaccard reaches 0.36 (Fibroblasts vs. Myeloid cells) while edge Jaccard stays below 0.09: cell types share hubs but rewire connections.

Part 5 repeats the GGM fit per-niche rather than per-class. Pairwise edge Jaccard between niches correlates moderately with composition cosine (Spearman $\rho = 0.615$), but the cluster-level family structure of Part 2 is not recovered (Adjusted Rand Index 0.25 for edges, 0.05 for hubs, vs. 0.80 for composition itself). Patient cohort overlap explains part of the structure ($\rho = 0.42$) but composition is the stronger driver. Niche 17, the B-cell + T-cell niche, emerges independently as a network outlier in the edge-similarity dendrogram, consistent with the distinctive microenvironmental gene programme of T cells identified in Part 4.

The most consistent finding across all five Parts is the structural distinctness of T cells, flagged by four independent analyses (boundary topology in Part 1, niche absence in Part 2, classification difficulty in Part 3, gene-network disjointness in Part 4). The unifying explanation is biological: T cells in tumour tissue are diffuse infiltrating populations with no consistent spatial neighbourhood, except in the TLS-like Niche 17 where they coexist with B cells in organised aggregates. The most generalisable methodological finding is the dominance of the spatial graph over per-cell modalities for the supervised task, and the convergence of cell-type and niche-level gene networks on a common "shared hubs, rewired connections" architecture.

Contents

1	Introduction	3
2	Spatial graph topology	5
3	Niche detection	10
4	Multi-modal GNN cell typing	14
5	Per-class gene networks	19
6	Per-niche gene networks	26
7	Discussion	31
8	Methods	35
9	References	39

1 Introduction

High-resolution spatial transcriptomics platforms such as 10x Genomics Visium HD now resolve gene expression at near single-cell resolution while preserving spatial coordinates and an aligned H&E histological image [1]. The resulting datasets are intrinsically network-shaped: every cell has neighbours, and those neighbours define a microenvironment that conditions its biology. Treating the data as a network — with cells as nodes and spatial proximity as edges — is not a methodological flourish but the natural framing of the object, and the lens through which I conduct every analysis in this report.

The dataset I work with comprises 8,065,561 individually segmented cells from a colorectal-cancer cohort of 56 patients (two samples per patient, 112 samples in total). Each cell carries a 16,497-dimensional gene expression vector, a 1280-dimensional H&E image embedding produced by the UNI foundation model [2], four morphological scalar features, and a categorical label drawn from 8 cell-type classes that span the tumour microenvironment: B cells (Class 0), Endothelial cells (Class 1), Fibroblasts (Class 2), Myeloid cells (Class 3), Normal epithelium (Class 4), SMC (Class 5), T cells (Class 6), and Tumor cells (Class 7). The cohort exhibits a $7.9\times$ class imbalance, with Tumor cells and B cells over-represented and T cells and Endothelial cells under-represented. Spatial adjacency is encoded as a k -nearest-neighbour graph with $k = 8$, capped by a per-sample radius threshold. Figure 1 illustrates the data-to-graph mapping on a representative sample: a tissue section becomes a point cloud coloured by cell type (panel a), zoomed regions reveal local cellular organisation (b), and the close-up exposes the KNN edges that connect each cell to its eight nearest neighbours (c).

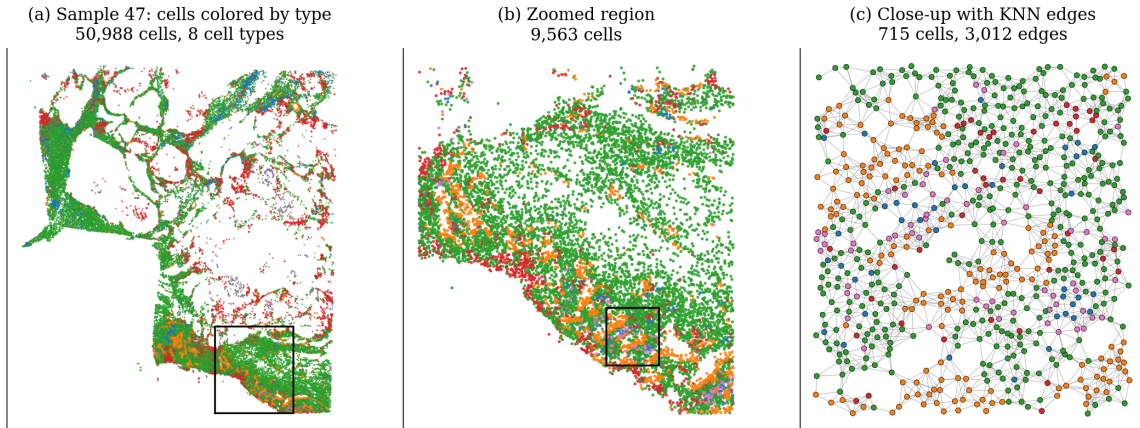


Figure 1: From spatial transcriptomics data to a graph. (a) Sample 47 with all 50,988 cells coloured by cell type; the black rectangle marks the region zoomed in (b). (b) A 9,563-cell region of the same sample; the inner rectangle marks the close-up shown in (c). (c) A 715-cell close-up with 3,012 KNN edges drawn as grey lines. Edges follow spatial proximity rather than cell-type identity: a cell is connected to its eight nearest neighbours regardless of class.

I organise the project into five Parts, each posing a specific question on the cohort. Part 1 asks what the spatial graph topology captures across the 112 samples: which metrics are consistent, which separate samples, which reflect biology rather than artefacts. Part 2 asks whether the tissue organises into recurrent microenvironments by detecting Leiden communities in each sample and aligning them across patients into a stable set of niches. Part 3 asks whether the spatial graph improves cell-type classification beyond per-cell features, evaluated through a five-rung ablation that progressively adds modalities (HVG expression, H&E embedding, morphology) and architectures (logistic regression, MLP, GraphSAGE, GAT). Part 4 asks whether each cell type carries its own gene co-expression network by fitting Gaussian Graphical Models on 378 broadly variable genes shared across classes. Part 5 repeats the same fitting per-niche rather

than per-class, then asks whether the resulting networks recover the composition-based niche families from Part 2.

Four findings stand out from the analysis. First, local graph topology is remarkably consistent across the cohort (mean degree near 8.2 for most samples), while a global metric — the fraction of cells in the largest connected component — splits the samples into two distinct modes, identifying compact and fragmented tissue architectures. Second, twenty niches recur stably across the cohort: 4 are universal (present in at least 85% of patients), 11 common, 4 selective, and only one (a pure-SMC niche) is rare, appearing in just 11 patients. The 20 niches partition asymmetrically across the eight cell types: Tumor cells, B cells, and Fibroblasts each host four niches, while T cells host none. Third, the spatial graph improves validation macro- F_1 by 6.5 percentage points over a strong per-cell baseline (0.6350 vs. 0.5704), with a final test macro- F_1 of 0.6089, while adding the H&E embedding, morphology features, or graph attention on top of GraphSAGE yields no further gain. Fourth, T cells emerge as the structurally distinctive class on every measure (topology, niche assignment, classification, gene network), with a unifying biological explanation: T cells in tumour tissue are diffuse infiltrating populations with no consistent spatial neighbourhood. Their one exception is Niche 17, a B-cell-dominant niche where T cells coexist with B cells at substantial proportions — a composition consistent with tertiary lymphoid structures, the only place in the niche landscape where T cells form an organised population. Across all 8 per-class and 20 per-niche gene networks, cell types and niches share dominant hubs (top-15 hub Jaccard up to 0.36 across classes, 0.58 across niches) but rewire the conditional-dependence connections between them (edge Jaccard at most 0.09 and 0.12 respectively).

The remainder of this report follows the five-Part structure (§2–§6), with a synthesis discussion in §7 and full methodological details in §8. Throughout, I refer to cell types by their biological labels (B cells, T cells, Tumor, etc.) in the discursive text, and add the numeric identifier (e.g. Class 6) at the first occurrence in each section and in cross-references to figures, which retain numeric c0–c7 labels for consistency with the cached analysis. Niches are referred to as Niche 0–Niche 19 with the dominant cell-type class annotated in parentheses in figures. Samples are indexed 0–111 and patients 0–55; the held-out test split is patients {0, 21, 26, 28, 34, 40, 45, 51}, fixed before any model training. One methodological qualification, important for reading Parts 5–6: at the near single-cell resolution of Visium HD, mRNA from a target cell and its immediate spatial neighbours is difficult to disambiguate, and per-cell gene expression vectors carry a non-trivial component of neighbourhood signal. The implications of this diffusion contamination for the gene-network analyses are discussed in detail in Part 4 and in §7.

2 Spatial graph topology

The spatial k -nearest-neighbour graph is the substrate of every analysis that follows: niche detection in Part 2 operates on it, the GNN in Part 3 consumes it through neighbour sampling, and the per-class and per-niche networks in Parts 5–6 are downstream of the niches it induces. The first question, therefore, is what this graph captures across the 112 samples of the cohort, and to what extent its topology reflects biology rather than artefacts of tissue handling or segmentation.

I summarise each sample with a free-tier of eight cheap-to-compute metrics — mean degree, fraction of isolated nodes, fraction of cells in the largest connected component, number of components of size ≥ 100 , global transitivity, mean local clustering, degree assortativity, and cell-type assortativity — and an expensive tier of vertex-level statistics (closeness, betweenness) computed on a subset of ten samples. The free tier is fast enough to run on the full cohort: the resulting parquet table has 112 rows and 22 columns and is loaded once at the start of every downstream notebook. Two samples (#87 and #107) emerge as topological anomalies during the free-tier pass and are excluded from subsequent cohort-level analysis, leaving 110 well-behaved samples.

Local topology is consistent across the cohort

The first observation is that the local statistics of the graph vary very little across patients. Figure 2 shows the distribution of mean degree per sample. After excluding the two anomalies, 108 of 110 samples concentrate between mean degree 6 and 9, with a median of 8.24 and a tight, single-peaked shape. This is essentially what one would expect from a KNN construction with $k = 8$ followed by symmetrisation and a radius cap: every sample sits close to that ceiling, with deviation driven by how often the radius cap discards far-away neighbours in sparse regions. Local transitivity is similarly uniform across the cohort (mean ≈ 0.53 , low spread), and the mean local clustering coefficient varies in a narrow band.

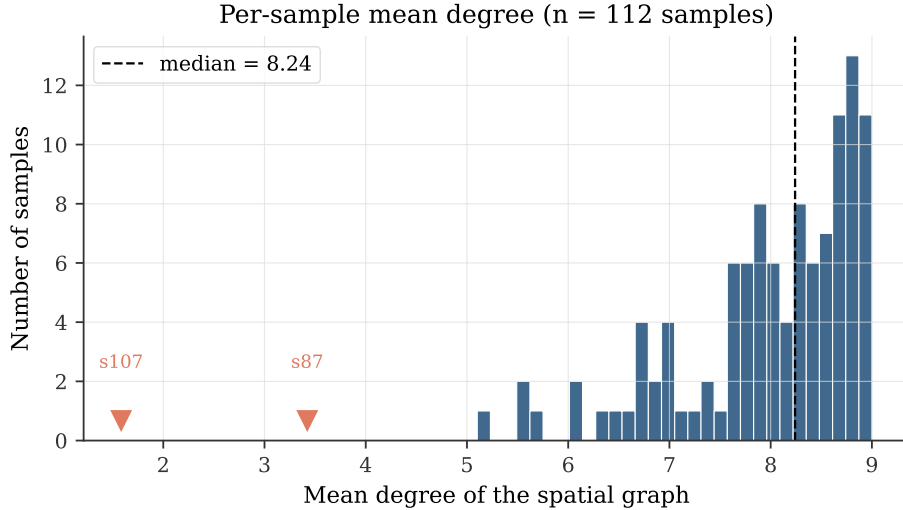


Figure 2: Per-sample mean degree of the spatial graph, across the 112 samples of the cohort. The two anomalies (#87, #107) sit far in the left tail at mean degree below 4 and are excluded from cohort-level analyses. The remaining 110 samples concentrate tightly around the median (8.24).

Global connectivity is bimodal

Local consistency does not extend to global connectivity. The fraction of cells belonging to the single largest connected component — a global statistic about how the tissue holds together — is sharply bimodal across the cohort (Figure 3). Roughly 17 samples have a largest-component fraction near 0.95: their tissue is essentially a single contiguous piece, what I call *compact architecture*. A second group of samples, with largest-component fraction in the $[0.25, 0.45]$ range, has its cells split across many roughly equal components; their tissue holds tendrill-like fragments. I refer to these as *fragmented architecture*. The middle of the histogram is populated but sparse, suggesting that the two modes are not extremes of a continuum but distinct configurations.

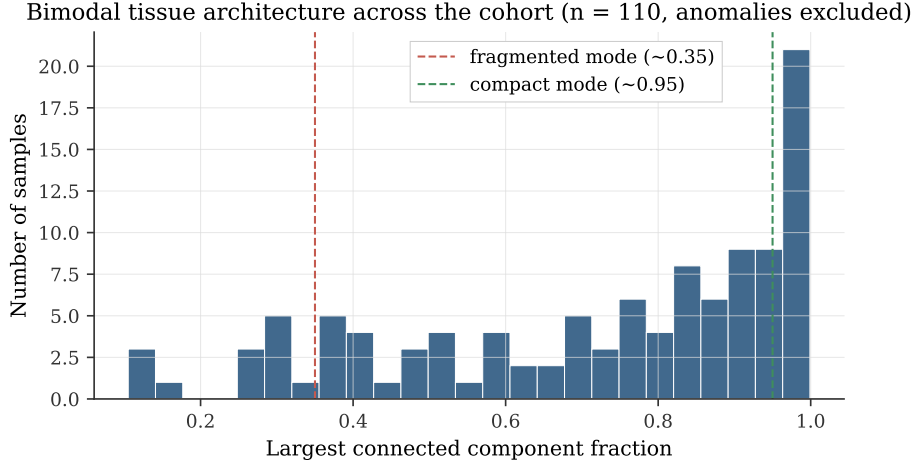


Figure 3: Histogram of the largest connected component fraction across 110 samples (anomalies excluded). The bimodal shape separates compact tissue (~ 0.95) from fragmented tissue (~ 0.35); the two modes are not extremes of a continuum but distinct architectures.

This is an important finding for downstream work: many of the analyses in Parts 3–6 subsample 5000 cells from a tissue region, and they will behave differently on compact and fragmented samples. I revisit this point in Part 2, where niche assignment recovers about 97% of cells in compact samples but only $\sim 55\%$ in fragmented ones (the unassigned cells live in many small Leiden communities that do not aggregate into niches).

I also checked whether the architecture is patient-consistent — whether the two samples of each patient land in the same mode. They do not: the Spearman correlation of largest-component fraction between paired samples of the same patient is only $r \approx 0.04$. Tissue architecture is therefore a per-sample property, not a per-donor one. This rules out the simplest interpretation (some patients have "compact tissue" by biological default) and points toward sample-level handling factors as the more plausible source of fragmentation.

Topology correlates with cell-type composition

Topology statistics are not just artefacts of the graph construction; they correlate with what is in the sample. Figure 4 shows the Spearman correlation between the eight topology metrics and the per-sample fraction of each cell-type class, computed across the 110 non-anomaly samples.

Two features of this heatmap are worth highlighting. First, SMC (Class 5) stands out as the dominant driver of topological variation across the cohort. Samples rich in SMC have lower mean degree ($\rho = -0.58$), lower clustering ($\rho = -0.60$), more isolated cells ($\rho = +0.51$), and more components of size ≥ 100 ($\rho = +0.64$). In other words, SMC marks sparse, fragmented regions of tissue — consistent with the perivascular and connective localisation of smooth

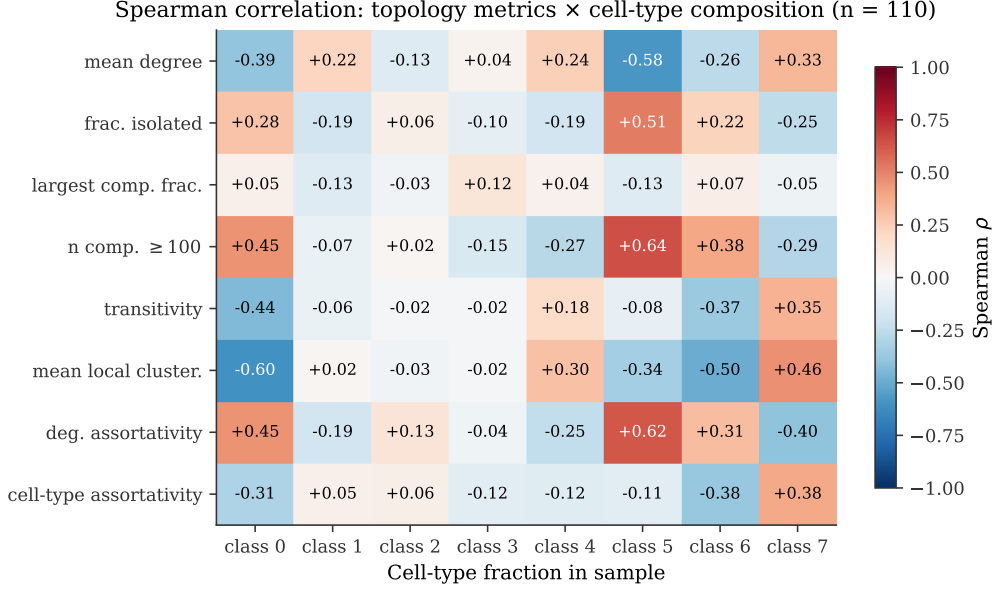


Figure 4: Spearman correlation between eight topology metrics (rows) and the per-sample fraction of each cell-type class (columns), across the 110 non-anomaly samples. SMC (Class 5) dominates the structure: it strongly anti-correlates with local density ($\rho = -0.58$ vs. mean degree, $\rho = -0.60$ vs. mean local clustering) and positively correlates with fragmentation indices ($\rho = +0.51$ vs. isolated fraction, $\rho = +0.64$ vs. number of components ≥ 100).

muscle cells, which occupy lower-density spatial niches than the dense epithelial or stromal compartments.

Second, B cells (Class 0) and T cells (Class 6) show an opposite pattern: they correlate negatively with sparsity ($\rho = -0.39$ and -0.26 for mean degree, $\rho = -0.50$ and -0.37 for mean local clustering) and positively with compactness. T cells also negatively correlate with degree assortativity ($\rho = -0.40$): they tend to sit at the boundary between high-degree and low-degree regions. This is the first cross-Part signal of a finding that will recur: T cells in tumour tissue are infiltrating populations that occupy boundary spatial contexts rather than homogeneous neighbourhoods. Part 2 will confirm that they never dominate any niche, Part 3 that they are the worst-classified cell type, and Part 4 that their gene network has no overlap with any other class.

The training mask reflects local density

The dataset provides a binary `train_mask` that indicates which cells were used for model training in the original benchmarking. Figure 5a shows that the per-sample fraction of cells retained by this mask is essentially a function of local density: Spearman $\rho = -0.91$ with mean degree. Dense samples have less of their cells in the training mask; sparse samples have more. This is a strong constraint to keep in mind for Part 3, where any model trained on the mask is implicitly trained on a density-biased subset of the cohort.

Betweenness reveals tissue skeletons

The cheap-tier metrics are sample-level scalars. To probe the per-cell structure I computed betweenness centrality on the giant connected component of ten representative samples, using Brandes–Pich source/target sampling [3] (full betweenness on a 42,000-node graph is computationally prohibitive). The result is unexpectedly visual: in every sample, the top 1% of cells by betweenness form a connected, branching pattern that traces the structural backbone of

the tissue. Figure 5b shows sample 2, where 425 cells of the 42,423-cell giant component are highlighted; the same pattern reproduces qualitatively on the other expensive-tier samples.

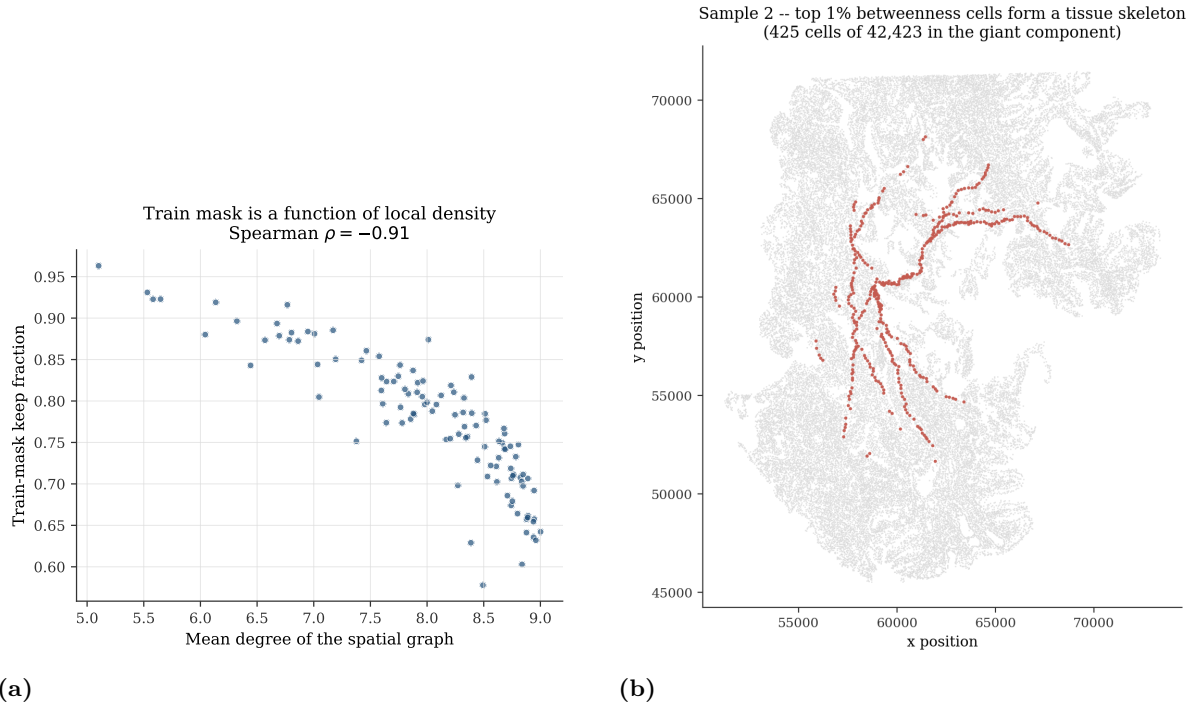


Figure 5: Beyond cohort-level metrics: per-sample and per-cell observations. **(a)** Per-sample train-mask keep fraction against mean degree, across 110 non-anomaly samples. The mask is almost deterministically anti-correlated with density ($\rho = -0.91$): dense samples have a smaller fraction of their cells in the training mask. Any classifier trained on the mask is implicitly trained on a density-biased subset of the cohort. **(b)** Sample 2 spatial layout, with the top 1% of cells by betweenness centrality (425 of 42,423) shown in red against the remainder of the giant connected component in grey. The high-betweenness cells form a connected, branching skeleton rather than a uniform spatial scatter: they sit at the bottlenecks between dense tissue blocks and act as graph-theoretic bridges between regions.

I checked whether betweenness-high cells are enriched for any particular cell-type class. Across the expensive-tier samples, no consistent enrichment emerges: the skeleton picks up whichever cell type happens to populate the inter-block bridges in that sample. The structural role is graph-defined, not biology-defined. Closeness centrality, in contrast, behaves essentially as an inverse Euclidean distance from the centroid of the sample ($r = 0.81$); it adds little information beyond geometry, and I drop it from subsequent analyses.

Anomaly handling

Two samples (#87 and #107) trip the anomaly check during the free-tier pass: their mean degree is below 4, fraction of isolated nodes is above 0.35, and largest-component fraction is below 0.1. Figure 6 compares one of them, sample #87, against a representative normal sample (#2): at the same absolute spatial scale, the anomalous sample displays a cell population that is both sparser and distributed over a much larger spatial extent. Quantitatively, sample #2 covers a $22\,000 \times 26\,000$ -unit region with 43,023 cells and a median nearest-neighbour distance of 52 units; sample #87 covers a $54\,000 \times 60\,000$ -unit region with only 24,268 cells and a median nearest-neighbour distance of 92 units. The 99th percentile of nearest-neighbour distance is even more telling: 107 units for sample #2 against 455 for sample #87. Because the spatial graph applies a cohort-wide radius cap calibrated on the typical inter-cell distance, the KNN edges in sample #87 are systematically rejected by the cap: cells exist in the data, but they

are too far apart to form a graph that resembles a connected tissue. The single small giant component visible at the bottom right of the panel is the only cluster dense enough to survive the cap.

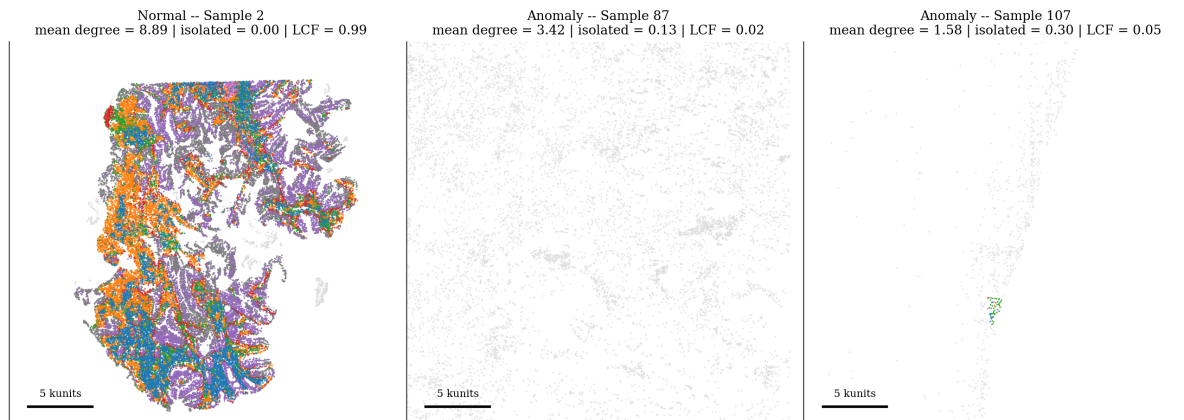


Figure 6: Anomaly handling: comparison of a representative normal sample (#2) against the two anomalous samples (#87, #107) at a common spatial scale ($30\,000 \times 30\,000$ units, centred on each sample’s centroid). Cells are coloured by cell type if they belong to the giant connected component of the spatial graph, light grey otherwise. The scalebar in each panel marks 5,000 units. Sample #2 fills its panel with densely connected coloured cells; samples #87 and #107 show cells distributed at a much lower density over a much larger spatial extent, with only a small giant component (the few coloured cells) surviving the radius cap on KNN edges.

I interpret these as tissue-handling or segmentation failures (necrosis, sample disaggregation, or segmentation that recovered only a subset of cells) rather than as a biologically meaningful sparse architecture; either way, the KNN-with-radius-cap construction is not a suitable substrate for their analysis. Both samples are excluded from all cohort-level analysis from this point forward. The remaining 110 samples form the working cohort used in Parts 3–6.

Summary. The spatial graph is locally uniform across the cohort but globally bimodal, separating compact from fragmented tissue architectures. Topology correlates with cell-type composition, with SMC as the dominant driver of sparsity. The training mask is almost entirely a function of local density. Per-cell betweenness identifies a tissue skeleton with no cell-type signature: it is a graph property of the geometry, not of the biology. These observations inform every downstream analysis, and constrain how robustly the niche detection in Part 2 and the GNN in Part 3 are expected to generalise across the architectural modes of the cohort.

3 Niche detection

Part 1 examined the spatial graph at the cohort level. Part 2 steps inside each sample to ask whether the tissue organises into recurrent microenvironments — spatial regions with a characteristic cell-type composition that reappear across patients. If such niches exist and are stable across the cohort, they provide a coarser, more biologically interpretable level of structure than individual cells, and a natural unit on which to anchor the downstream analyses in Parts 4–6.

Per-sample communities and cross-sample alignment

Within each sample I run the Leiden algorithm [4] with resolution parameter $\gamma = 0.5$ on the weighted giant-component subgraph, where edge weights derive from cell–cell expression similarity. Resolution 0.5 was selected after a small sweep on three representative samples: lower values produce 2–3 monolithic communities, higher values fragment into many small communities of ~ 30 cells. At $\gamma = 0.5$ the median sample yields 76 communities, with the full cohort summing to 11,496 non-anomaly communities across 110 samples.

Each community is summarised by its cell-type composition: an 8-dimensional vector giving the fraction of cells in that community belonging to each Class 0–Class 7. To align communities across samples I filter to those with at least 100 cells (9,457 communities of the 11,496) and apply hierarchical clustering with Ward linkage and Euclidean distance on the composition vectors. The number of clusters K is chosen by inspecting the dendrogram and the within-cluster homogeneity at several values of K ; $K = 20$ is the smallest value at which clusters remain compositionally pure (median intra-cluster cosine distance ≤ 0.15). The 20 clusters become the *niches* of the project, and every community is assigned to its niche centroid. Cells in communities below the 100-cell threshold are left unassigned; the assignment covers 77% of cohort cells overall.

The 20 niches form a coherent compositional landscape

The composition centroid of each niche, sorted by dominant cell-type class, is shown in Figure 7. The 20 niches partition across the eight cell types as follows: four are dominated by Tumor cells (Niche 0, Niche 1, Niche 2, Niche 4), four by B cells (Niche 16, Niche 17, Niche 18, Niche 19), four by Fibroblasts (Niche 3, Niche 9, Niche 10, Niche 12), three by Normal epithelium (Niche 5, Niche 6, Niche 7), two by Myeloid cells (Niche 8, Niche 13), two by SMC (Niche 14, Niche 15), one by Endothelial cells (Niche 11), and zero by T cells. The asymmetric distribution across cell types reflects both the underlying cohort composition (Tumor and B cells are abundant; T cells and Endothelial are rare) and the spatial homogeneity of each population (T cells are infiltrating and disperse, so they never accumulate into a niche of their own).

Several niches are remarkably pure (dominant fraction ≥ 0.75): Niche 0 is 95% Tumor, Niche 1 is 82% Tumor, Niche 5 is 88% Normal epithelium, Niche 9 is 76% Fibroblast, Niche 8 is 72% Myeloid, Niche 15 is 87% SMC, and Niche 16 is 79% B cells. These are the "characteristic" niches of their cell type — regions where one population is overwhelming and the local microenvironment is largely homogeneous.

Other niches are intentional mixtures that the hierarchical clustering refused to merge with the nearest pure neighbour, because the mixture itself recurs across patients with sufficient consistency to define its own niche. Niche 4 is 46% Tumor + 17% Fibroblast + 17% Myeloid (a tumour-stroma-inflammation interface niche); Niche 3 is 38% Fibroblast and 37% Tumor in roughly equal parts (a tumor-stroma border); Niche 14 is 42% SMC + 35% B cells (a vascular-immune interface, see niche 17 below); Niche 6 is 40% Normal epithelium + 35% B cells (an epithelial-immune boundary); Niche 13 is 27% Myeloid + 19% B cells + 26% Fibroblast (a

highly heterogeneous mixed niche).

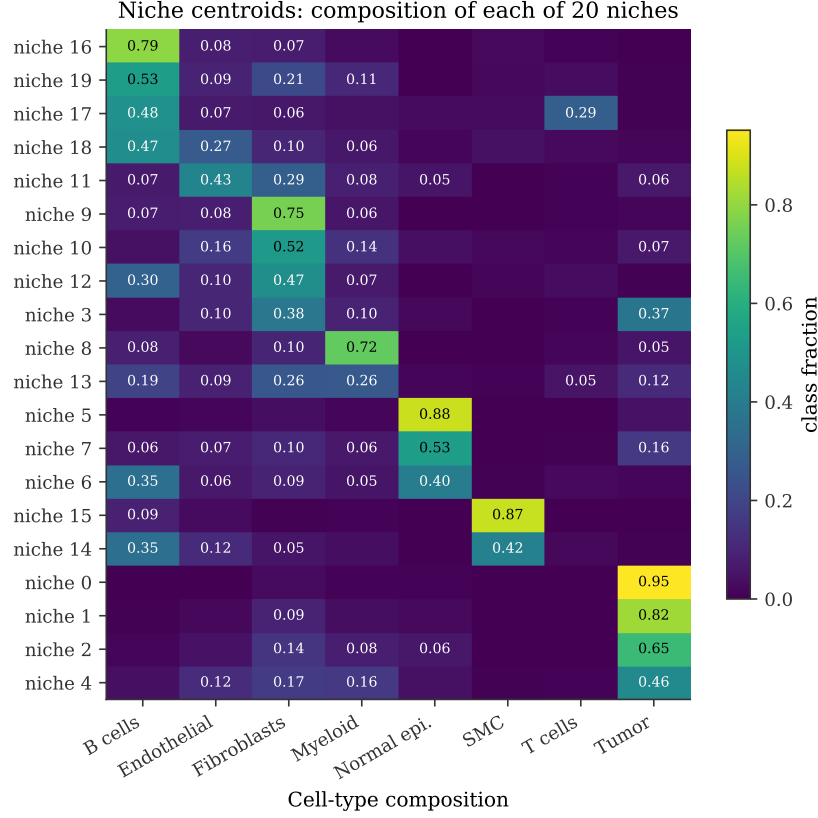


Figure 7: Composition centroids of the 20 niches, sorted by dominant cell-type class (primary) and dominant fraction descending (secondary, so the purest niches of each compositional family appear at the top of their block). Each row is a niche; columns give the fraction of cells in that niche belonging to each cell type, with biological labels. Only values ≥ 0.05 are annotated for legibility. T cells never dominate any niche but appear as a 29% secondary minority in Niche 17, the B-cell + T-cell niche likely corresponding to tertiary lymphoid structures.

T cells never dominate a niche — but cluster with B cells where they appear

A striking feature of this landscape is that T cells (Class 6) never dominate any of the 20 niches. In 18 of 20 niches, T cells contribute less than 5% of the composition. Niche 13 sits just above the threshold at 5.1%. The only niche where T cells form a substantial population is Niche 17, where they make up 29% of the composition alongside the dominant B cells (48%).

The composition of Niche 17 (B cells 48% + T cells 29%, with Endothelial cells at 7% as the next contributor) is consistent with the morphology of tertiary lymphoid structures (TLS) in tumour tissue: organised aggregates of B and T cells, typically perivascular, that form as part of the anti-tumour immune response. This interpretation will become important in Part 4, where the per-class GGM of B cells (Class 0) recovers a smooth-muscle hub signature consistent with the perivascular localisation of B cells in TLS.

The structural distinctness of T cells will recur in every subsequent Part: they are the worst-classified class in the GNN of Part 3, they form the most isolated network in the per-class GGM of Part 4, and they are the dominant secondary in the only niche that lies apart from its compositional family in the network-based dendrogram of Part 5. All four signatures point to the same underlying biology: T cells in tumour tissue are diffuse infiltrating populations with no consistent spatial neighbourhood.

Niches are spatially coherent regions

The niches were defined by composition, not by space. If the algorithm has captured something real, niche assignments should form spatially contiguous regions within each sample, not random scatter. Figure 8 shows two representative samples with cells coloured by their assigned niche. In both samples (sample 2, compact, 43,023 cells; sample 20, compact, 45,068 cells) the niches form coherent spatial blocks: large stretches of a single niche colour, with sharp boundaries at the interfaces. Unassigned cells (cells in Leiden communities below the 100-cell threshold) appear as occasional grey speckles in regions of low cell density.

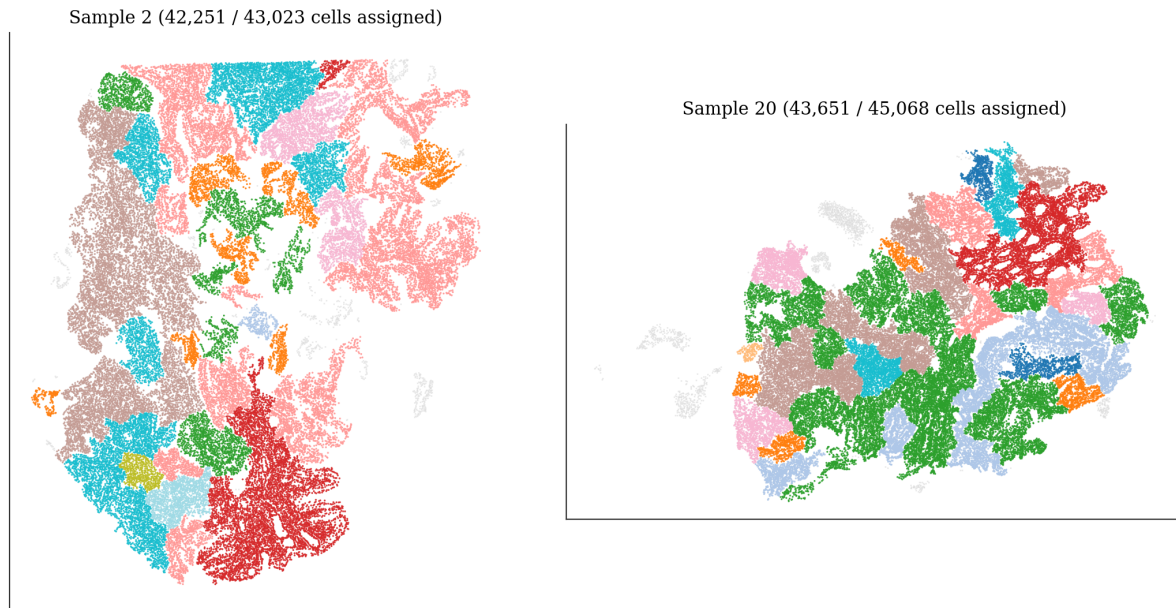


Figure 8: Spatial layout of two representative samples, with cells coloured by their assigned niche. The colour palette is shared between panels (`tab20`). Niches form coherent spatial blocks rather than random scatter, supporting the interpretation that they correspond to genuine tissue microenvironments rather than algorithmic artefacts. Unassigned cells appear as grey speckles, mostly in regions of lower cell density.

I checked sample 47 (a fragmented architecture, from Part 1) as a stress test: only 38% of its cells receive niche assignments, with most unassigned cells lying in the upper tendrils of the tissue. This is not a failure of niche detection but a structural consequence of fragmentation: the Leiden communities in fragmented samples are smaller, more rarely cross the 100-cell threshold, and therefore drop out of the alignment step. Compact samples like #20 and #2 in Figure 8 consistently exceed 95% assignment, while fragmented samples vary between 50% and 60%. I treat the niche analysis as well-defined on the cohort but with weaker resolution on the fragmented mode.

Niche recurrence partitions the niches into four bands

How widely a niche appears across the 56 patients is itself an informative property. For each niche, I count the number of distinct patients that contribute at least one cell to it; the distribution is shown in Figure 9. The 20 niches split into four recurrence bands:

- **Universal** (≥ 48 patients, $\geq 85\%$): four niches recur in nearly every patient. These are Niche 4 (53 patients, Tumor-mixed), Niche 2 (52, Tumor-dominant), Niche 1 (48, Tumor pure), and Niche 13 (48, Myeloid heterogeneous). The fact that three of the four are Tumor-dominant reflects the centrality of Tumor cells in the cohort: tumour tissue is the common substrate.

- **Common** ($28 \leq \text{patients} < 48$): eleven niches are present in at least half of patients but not all. They include the Tumor-pure Niche 0 (45 patients), the Fibroblast-pure Niche 9 (37), the various B-cell niches (Niche 19 at 43, Niche 16 at 39, Niche 18 at 38, Niche 17 at 31), and most of the Fibroblast and Normal-epithelium niches.
- **Selective** ($14 \leq \text{patients} < 28$): four niches recur in a quarter to half of patients. These tend to be the purer or more specialised niches: Niche 6 (25 patients, Normal-ep-mixed), Niche 14 (24, SMC-mixed), Niche 8 (23, Myeloid pure), Niche 5 (22, Normal-epithelium 88% pure).
- **Rare** (< 14 patients, $< 25\%$): one niche appears only in a small minority of the cohort. Niche 15, the only pure-SMC niche (87% SMC), recurs in 11 patients.

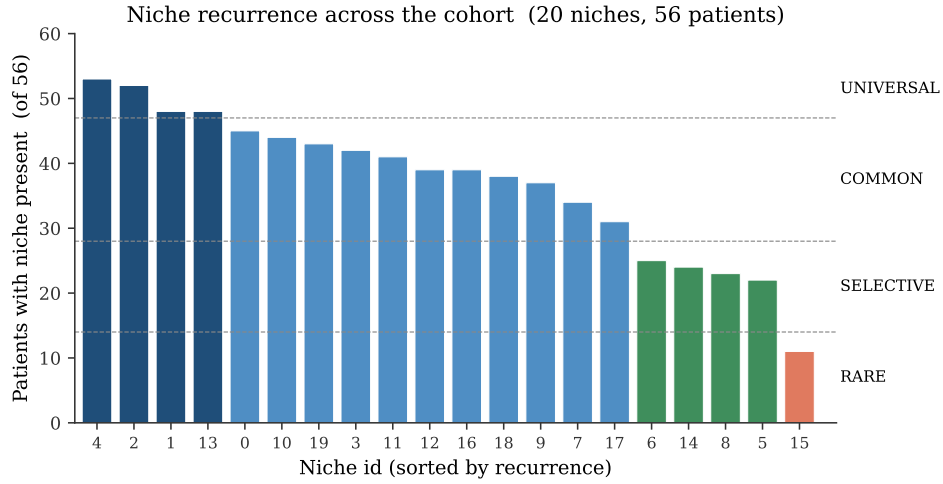


Figure 9: Number of patients containing each niche, sorted by recurrence. Niches partition into four bands: universal (dark blue), common (lighter blue), selective (green), and rare (orange). Reference lines mark the band thresholds at 85%, 50%, and 25% of the 56 patients. The single rare niche (Niche 15, pure SMC at 87%) appears in only 11 patients — a structural minority of the cohort.

Niche 15 is the most informative rare niche for what comes next: pure SMC, restricted to 11 patients, with the second-highest count of "silent" common genes in the per-niche GGM analysis of Part 5 (24 of 378 common genes have zero variance in this niche's subsample, second only to Niche 5 at 32). Its narrow patient base and specialised cell-type profile make it both biologically distinctive and statistically challenging to model.

Summary. Twenty stable niches emerge from per-sample Leiden communities aligned across the cohort by hierarchical clustering on composition. They cover 77% of cohort cells, form spatially coherent regions in compact samples, and split into four recurrence bands from universal to rare. Seven of the eight cell types host one to four niches; T cells never dominate any. Niche 17, the B-cell-dominant niche with 29% T-cell secondary, is the only place in the niche landscape where the two adaptive-immune cell types coexist in substantial proportions — a composition consistent with tertiary lymphoid structures. The 20 niches become the units of analysis for Parts 4 and 6.

4 Multi-modal GNN cell typing

Part 3 asks a different kind of question from the previous Parts. So far the analysis has been descriptive: the spatial graph captures certain topological patterns, and the niches it induces are stable and interpretable. Whether any of this structure is *predictive* of cell-type identity is a separate matter. Part 3 answers that question through a controlled five-rung ablation, where I progressively add modalities (HVG expression, H&E embedding, morphology) and model classes (logistic regression, MLP, GraphSAGE, GAT) and watch macro- F_1 track each addition.

Task and split

The supervised task is per-cell classification into the eight cell-type classes introduced in §1 (B cells, Endothelial, Fibroblasts, Myeloid cells, Normal epithelium, SMC, T cells, Tumor), with class imbalance of $7.9\times$ between the most and least frequent class. The patient-level split is fixed once at the beginning of the project and reused throughout: 40 patients for training, 8 for validation, 8 for testing. Validation macro- F_1 drives early stopping and rung selection. All models are trained on the cells flagged by the dataset’s `train_mask` field (which, as shown in Part 1, is itself a density-biased subset of the cohort).

For input features, the gene expression vector is first reduced from the full 16,497 genes to a 1000-dimensional highly variable gene (HVG) selection computed with the Seurat criterion [5, 6] on the training cells only. The 1280-dimensional UNI H&E embeddings and the four morphology features are added as separate modalities for the multi-modal rungs. Per-modality feature standardisation uses the training-cell statistics only.

The 5-rung ablation

The five rungs are designed to isolate the contribution of each modality and each architectural choice. **Rung 1** is a logistic regression on HVG features alone; it establishes the baseline of "per-cell expression with no neighbourhood information." **Rung 2** is a modality-fused MLP that concatenates HVG, H&E, and morphology through per-modality encoders before a shared classification head; it tests whether per-cell features alone, with all modalities, can saturate the task. **Rung 3** is a 2-layer GraphSAGE [7] on HVG features only; it isolates the contribution of the spatial graph by comparing against Rung 2 (multi-modal, no graph). **Rung 4** adds the H&E and morphology modalities to GraphSAGE; it tests whether modalities and graph compose. **Rung 5** replaces GraphSAGE with a 2-layer GAT [8] of 4 attention heads, again with all modalities; it tests whether attention over neighbours is useful beyond uniform neighbour aggregation. All GNN training uses neighbour sampling with batch size 2048 and early stopping with patience 3 on validation macro- F_1 .

Figure 10 summarises the validation macro- F_1 of the best epoch for each rung. The pattern is clean enough to extract three claims directly.

First, the spatial graph helps substantially. Rung 3 (GraphSAGE, HVG only) reaches macro- $F_1 = 0.6350$, a 6.5-percentage-point gain over the multi-modal MLP at Rung 2 (macro- $F_1 = 0.5704$). The graph signal therefore provides more information than the H&E embedding and morphology features combined.

Second, modalities do not compose with the graph. Rung 4 (GraphSAGE with all modalities) reaches macro- $F_1 = 0.6308$, slightly *below* the HVG-only Rung 3. Adding H&E and morphology on top of a graph model not only fails to help, it marginally hurts. One interpretation is that the modalities encode some of the same neighbourhood signal that GraphSAGE already extracts through aggregation, and the additional features act as a regularisation tax rather than a true information gain.

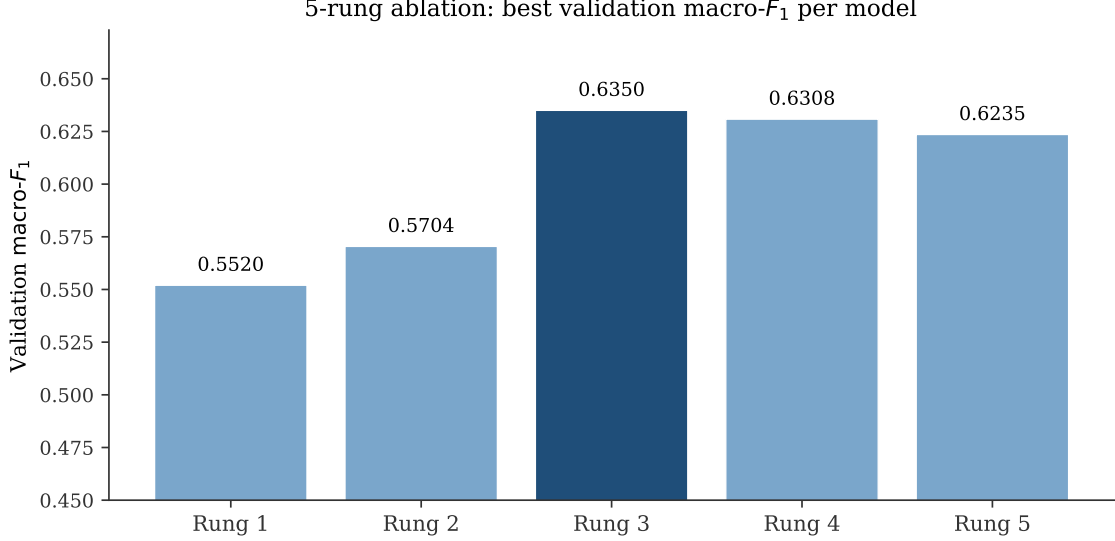


Figure 10: Validation macro- F_1 of the best epoch for each of the five rungs. The spatial graph (Rung 2 $\rightarrow$ Rung 3) drives a +6.5 percentage-point gain over a strong per-cell baseline. Adding modalities on top of GraphSAGE (Rung 4) does not improve over Rung 3, nor does replacing aggregation with attention (Rung 5). The winning rung (highlighted in dark blue) is Rung 3, GraphSAGE on HVG features alone.

Third, attention does not improve over uniform aggregation. Rung 5 (GAT, all modalities) reaches macro- $F_1 = 0.6235$, the worst of the three GNN rungs. The optimal cell-type prediction does not benefit from learned, edge-specific weighting of neighbours. Combined with the previous claim, this suggests that the right inductive bias for this task is a simple mean-aggregating GNN on HVG features — which is exactly Rung 3.

Modality masking on the multi-modal MLP

To verify that the gain in Rung 3 is graph-driven and not just an architectural side-effect, I ran a modality-masking analysis on Rung 2, the multi-modal MLP. Setting each modality’s encoder output to its mean (effectively removing the modality without changing model capacity) yields the following drops in validation macro- F_1 :

- HVG masked: macro- F_1 drops by -39 percentage points (catastrophic, as expected)
- H&E masked: macro- F_1 drops by -8.5 percentage points
- Morphology masked: macro- F_1 drops by -0.1 percentage points (negligible)

The H&E embedding contributes 8.5pp inside the multi-modal MLP — a real but modest signal — yet the same embedding contributes nothing when added on top of GraphSAGE. The graph and the H&E embedding therefore overlap in the information they provide about a cell’s identity. This is biologically plausible: H&E captures the local tissue architecture (which is exactly what the spatial graph propagates), so the two modalities are partially redundant.

Per-class progression and confusion structure

The aggregate validation F_1 hides substantial class-level variation. Figure 11 traces the per-class validation F_1 across the five rungs. Several patterns emerge:

Tumor cells (Class 7), the highest-performing class throughout, jump from $F_1 = 0.79$ at Rung 2 to 0.88 at Rung 3 — the largest single-rung gain in the figure, and the strongest evidence that GraphSAGE helps the abundant classes. Normal epithelium (Class 4) also jumps substantially

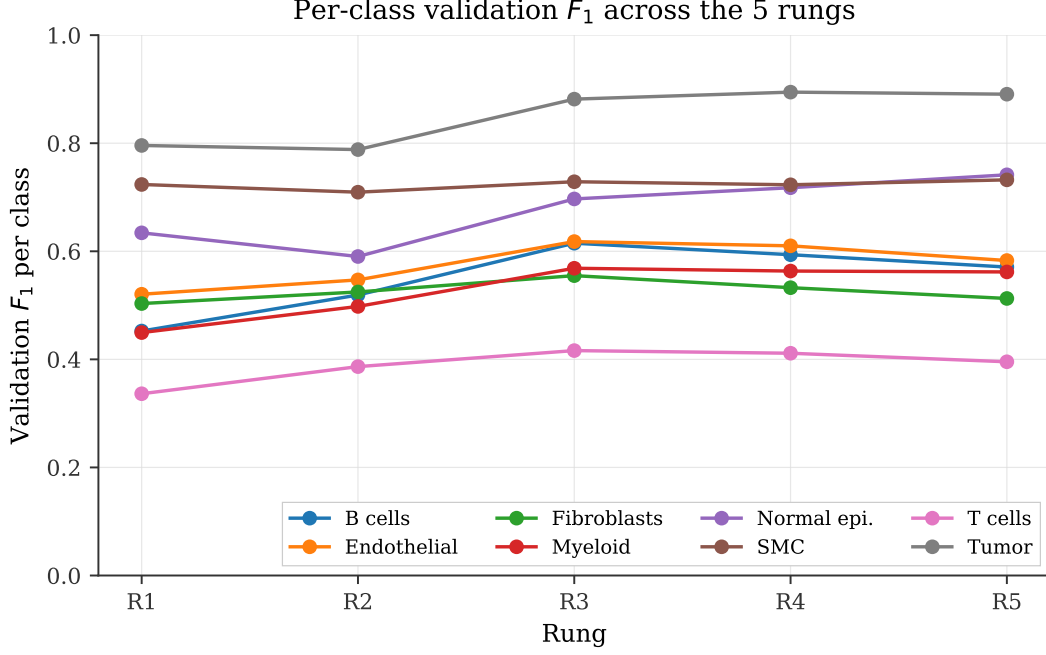


Figure 11: Per-class validation F_1 across the 5 rungs. Each line tracks one cell-type class. The graph (R2 $\rightarrow$ R3) produces large gains for B cells, Normal epithelium and Tumor cells; near-zero gains for SMC; and partial declines on R4–R5 for B cells and Fibroblasts.

(from 0.59 to 0.70). B cells (Class 0) jump from 0.52 to 0.62, then slowly decline through R4 and R5. SMC (Class 5), in contrast, sits near $F_1 \approx 0.72$ at all five rungs: per-cell features already saturate its classification, and the graph adds nothing. T cells (Class 6), the structurally distinctive class identified in Part 1 and Part 2, remain the worst-performing class throughout (peak $F_1 \approx 0.42$ at Rung 3), consistent with their dispersed, infiltrating spatial distribution: neighbourhood features cannot discriminate T cells effectively because their neighbourhood lacks a consistent signature.

Two classes regress on the multi-modal GNN rungs. B cells drop from 0.62 (R3) to 0.56 (R5), and Fibroblasts drop from 0.55 (R3) to 0.51 (R5). These are the two classes whose performance is most hurt by adding modalities on top of GraphSAGE — the cases where the redundant H&E/morphology signal acts as net noise.

Figure 12 shows the row-normalised confusion matrix of Rung 3 on the test set. The diagonal is highest for SMC (94% recall), Normal epithelium (89%), and T cells (84%) — the latter despite its low F_1 , because precision rather than recall is the limiting factor for T cells. The dominant off-diagonals are B cells $\rightarrow$ T cells (14%), Tumor $\rightarrow$ Normal epithelium (25%), and Fibroblasts $\rightarrow$ Myeloid cells (17%); these are the systematic mistakes the model makes, and they correspond to the niche-mixing patterns of Part 2. The Tumor $\rightarrow$ Normal-epithelium confusion is consistent with Niche 7 (Normal epithelium 53% + Tumor 16%), the tumor-epithelial border niche, where Tumor cells abut adjacent normal tissue and become locally indistinguishable from it. The Fibroblasts $\rightarrow$ Myeloid confusion is consistent with the inflammatory cluster of niches Niche 9–Niche 13, where Fibroblasts and Myeloid cells coexist at substantial proportions. The B cells $\rightarrow$ T cells confusion is the signature of Niche 17, the TLS-like niche.

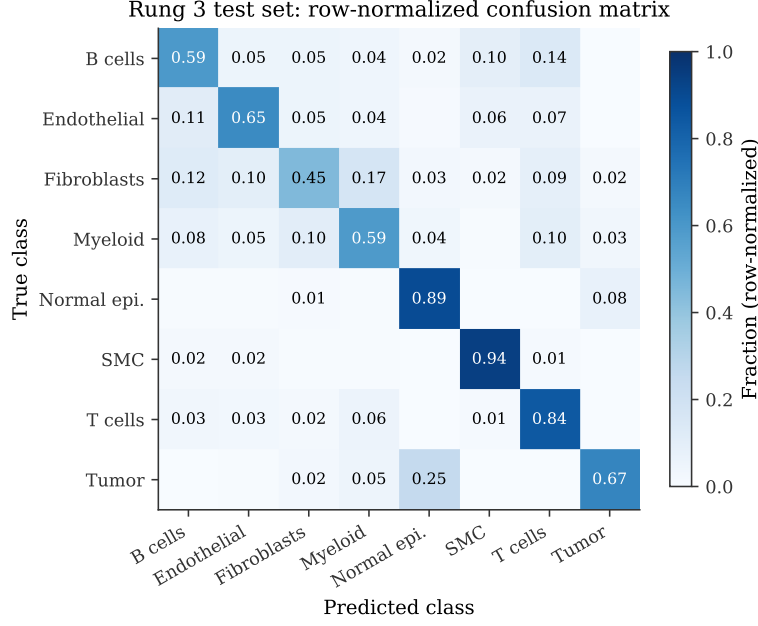


Figure 12: Row-normalised confusion matrix of Rung 3 on the test set. Diagonal entries give per-class recall. The strongest off-diagonal confusions (B cells $\rightarrow$ T cells, Tumor $\rightarrow$ Normal epithelium, Fibroblasts $\rightarrow$ Myeloid) correspond to the niche-mixing patterns identified in Part 2.

Training dynamics

The training curves of Rung 3 (Figure 13) reveal that GraphSAGE converges very quickly on this task. The training loss decreases monotonically from 0.79 to 0.72 over six epochs, but the validation macro- F_1 oscillates in the very narrow range $[0.627, 0.635]$ from epoch 1 onwards. The model reaches near-optimal validation performance after a single epoch; subsequent training refines the train loss without improving validation. Early stopping with patience 3 terminates training at epoch 6, picking the best validation epoch (epoch 3) as the checkpoint. This pattern is consistent with the strong saturation observed in the per-class progression: cell-type identity in this dataset is largely a function of immediate-neighbourhood features, and GraphSAGE finds the relevant signal almost immediately.

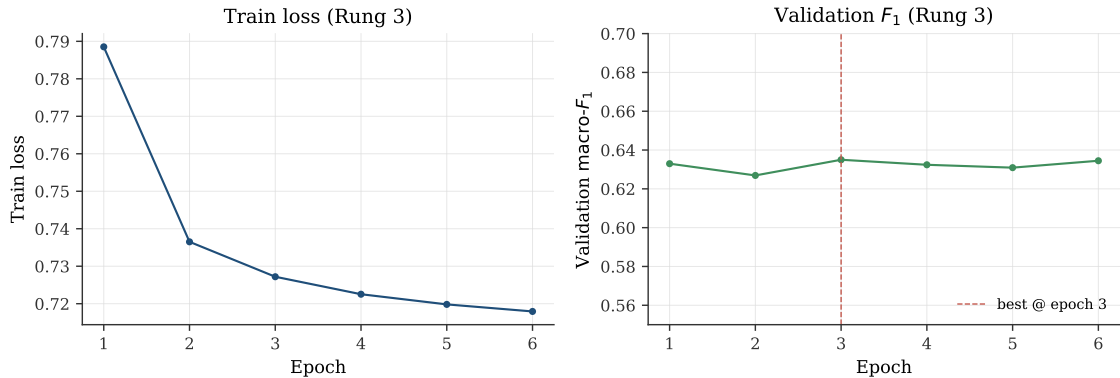


Figure 13: Training curves of Rung 3. Train loss (left) decreases monotonically, while validation macro- F_1 (right) plateaus near 0.633 after the first epoch. The dashed line marks the best validation epoch (epoch 3). The model converges very quickly on this task.

Test set evaluation and per-niche breakdown

Selecting Rung 3 as the winning configuration on the validation set, I run a single evaluation on the held-out 8-patient test set. The test macro- F_1 is macro- $F_1 = 0.6089$, a -2.6pp gap from the validation F_1 . This is small but real, and it sets the realistic upper bound on what this approach can achieve on unseen patients. Per-patient test macro- F_1 ranges from 0.473 (patient 0) to 0.625, with mean 0.547 and standard deviation 0.042; the patient-level variation is small, suggesting the model generalises consistently rather than relying on a few easy patients.

The more informative breakdown is per-niche (Figure 14). Test macro- F_1 by niche varies from ~ 0.24 (Niche 0, pure Tumor) to ~ 0.85 (Niche 12, Fibroblasts + B cells mix). The worst-performing niches are the small, niche-pure ones in the test patients — Niche 0, Niche 15, Niche 1, Niche 14 all score below the median — while the heterogeneous mixed niches (Niche 12, Niche 19, Niche 10, Niche 18) score above 0.70. The median per-niche test F_1 is only 0.449, well below the aggregate test F_1 of 0.609, because the large heterogeneous niches dominate the cell count and pull the aggregate up.

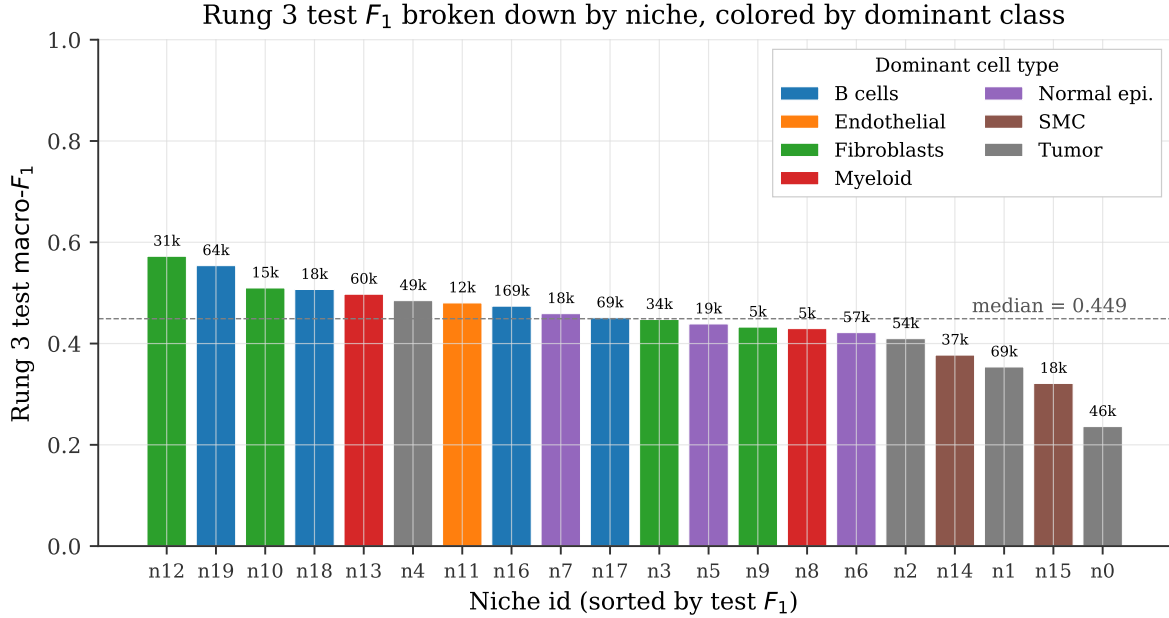


Figure 14: Rung 3 test macro- F_1 broken down by niche, sorted descending. Each bar is coloured by the dominant cell-type class of the niche. Cell counts are annotated above each bar. The median per-niche test F_1 (0.449) is well below the aggregate test F_1 (0.609), reflecting that large heterogeneous niches dominate the aggregate. Small homogeneous niches (Niche 0, 15, 1) are the most difficult.

This is an important nuance for downstream use of the model: the macro- F_1 of 0.609 is a fair summary of the model’s quality on a per-cell basis, but it masks substantial heterogeneity across spatial regions. A niche-pure region in a test patient is much harder than a heterogeneous one, because per-cell features in a pure region give the model less to distinguish.

Summary. A 2-layer GraphSAGE on HVG features achieves test macro- F_1 of 0.6089, an absolute improvement of $\sim 5\text{pp}$ over a strong multi-modal MLP baseline that has no access to the graph. Adding the H&E embedding or graph attention on top of GraphSAGE yields no further gain. The spatial graph is the single most informative architectural choice for this task, and one that subsumes a substantial fraction of the information carried by the H&E modality. T cells remain the structurally hardest class in every configuration, consistent with their dispersed, infiltrating spatial distribution and their boundary topological role from Part 1–Part 2.

5 Per-class gene networks

The first three Parts treat each cell as a node in a spatial graph. Part 4 flips this around: each cell-type class is treated as a population, and the structure I want to recover is the network of conditional dependencies between genes within that population. The tool of choice is the Gaussian Graphical Model (GGM), fit by the Graphical Lasso [9]. A non-zero entry Θ_{ij} in the sparse inverse covariance matrix indicates that genes i and j are conditionally dependent given all the other genes; intuitively, an edge in the gene network represents a direct co-expression relationship that survives controlling for everything else. The question of Part 4 is: do the 8 cell-type classes carry distinct gene networks, and to what extent do those networks share structure?

A second methodological caveat concerns the distributional assumption of the Graphical Lasso. The estimator assumes that the gene expression vectors are multivariate Gaussian, an assumption that spatial-transcriptomics counts violate: log-normalised counts retain zero-inflation and right-skew, especially for genes with cell-type-restricted expression, and the variance grows with the mean rather than being constant. I attenuate this by restricting to a common-gene subset of broadly variable genes (see below), but the violation is real. A principled semiparametric alternative is the nonparanormal SKEPTIC estimator of Liu et al. [10], which replaces Pearson correlations with Kendall’s τ or Spearman’s ρ and is consistent under arbitrary continuous marginal transformations. I opted for the standard Graphical Lasso because the conclusions of Part 4 are qualitative (hub identity, shared-hubs vs. rewired-connections) and remain stable across bootstrap resamples; a nonparanormal refit would be a natural validation step but is not pursued in this report.

Before presenting results, a methodological note is necessary. The Visium HD platform resolves expression at near single-cell scale, but at this resolution mRNA from a target cell and its immediate spatial neighbours becomes difficult to disambiguate: a non-trivial fraction of reads assigned to a cell originates from cells within ~ 1 to 2 neighbours of it. This phenomenon, usually called *diffusion contamination* (or molecular bleed-through), means that the expression vector of a cell is partly a reflection of its spatial neighbourhood. Any conditional-dependence network fit on per-cell expression will therefore reflect a mix of cell-intrinsic regulation and the gene programmes of spatially adjacent cells. The strength of this effect varies across cell types: for cells that sit in a homogeneous spatial context the diffusion reinforces the cell-intrinsic signal, while for cells that sit at the interface between two distinct compartments (or for low-abundance cells whose own transcripts are outnumbered by neighbour transcripts) the diffusion may dominate. I refer to the resulting structure as the cell type’s *microenvironmental gene programme*, with the understanding that this captures both intrinsic regulation and neighbourhood signal in proportions that vary by class. This is not a methodological flaw of Part 4 but a property of the data; I revisit its implications explicitly in the Discussion.

The 378 common-gene subset and the choice of α

A GGM on 16,497 genes would be both computationally expensive and uninterpretable: with order 10^8 candidate edges per class and a sample size of $\sim 10^5$ cells, the model would be heavily over-parameterised. Instead, I restrict the analysis to a common-gene subset: the genes among the 1000 HVGs from Part 3 that have standard deviation above 0.05 in *every* one of the 8 cell-type classes. This is 378 genes. Restricting to genes that are broadly variable across classes ensures that any class-specific structure cannot trivially be attributed to a gene being silent in some classes and expressed in others; the differences that survive are differences in network topology.

The regularisation parameter α controls sparsity: higher α gives sparser networks. I selected $\alpha =$

0.07 by sweeping over $\{0.05, 0.07, 0.097, 0.136, \dots, 1.0\}$ on two contrasting classes (SMC, with the most distinct expression pattern; Tumor cells, with the largest expected gene programme) and targeting roughly 1% edge density. At $\alpha = 0.07$ the eight networks have 494–980 edges out of $\binom{378}{2} = 71,253$ possible, corresponding to 0.7%–1.4% density. The full sweep showed that the qualitative findings (hub identity, cross-class Jaccards) are stable in the range $\alpha \in [0.05, 0.1]$, so the choice of 0.07 is not delicate. Numerical caveats: Endothelial cells, Fibroblasts, and T cells did not reach the Graphical Lasso convergence tolerance within 200 iterations at $\alpha = 0.07$; their precision matrices are still well-defined but the dual gap is non-zero. I report results from the best iterate. For each class I subsample 5000 cells from the training pool (patient-stratified) before fitting.

Network topology varies smoothly across classes

The eight fitted networks all have right-skewed, long-tailed degree distributions (Figure 15). Most genes have degree below 10; a small number of hub genes have degree above 20 and concentrate the network’s connectivity. This is the typical "scale-free flavour" of biological networks, though I do not claim a power-law fit on degree distributions of this size. The class-to-class variation in degree distribution is modest in shape but substantial in scale: Myeloid cells (Class 3) and Tumor (Class 7) have the longest tails (max degree above 35), T cells (Class 6) have the shortest (max degree ~ 18), and Normal epithelium (Class 4) and T cells (Class 6) have the narrowest distributions overall. The numerical edge counts span a factor of 2 across classes (494 to 980), reflecting differences in how much regulatory structure the Graphical Lasso recovers from each class’s expression pattern.

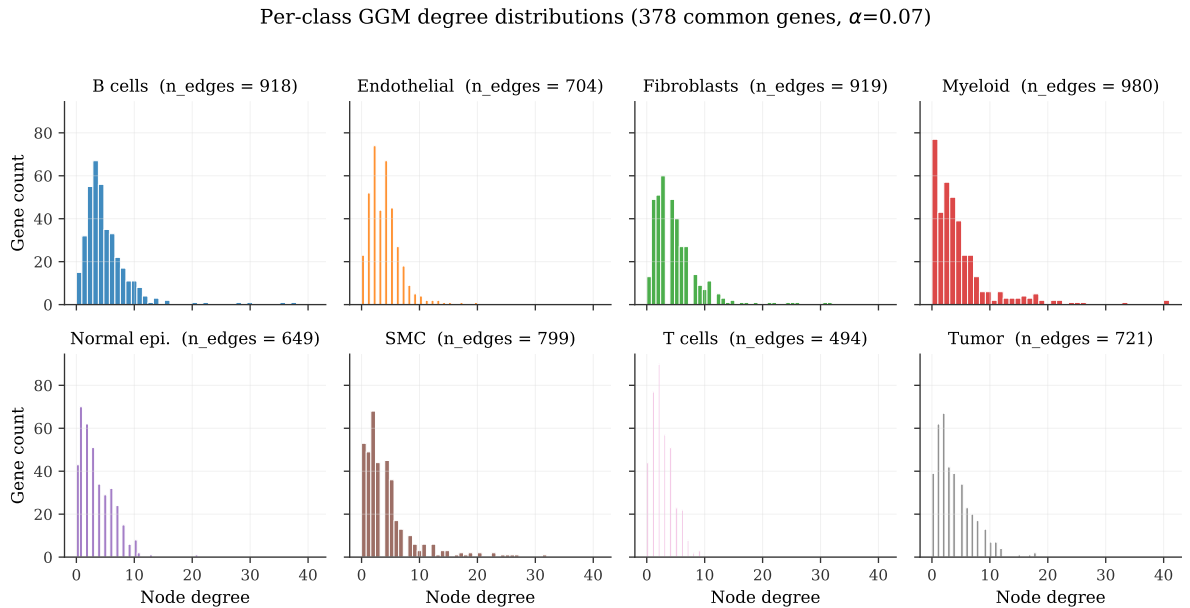


Figure 15: Degree distributions of the eight per-class GGMs at $\alpha = 0.07$ on the 378 common genes. All distributions are right-skewed with long tails (a small number of hubs concentrate connectivity). T cells (Class 6) have the sparsest network ($n_{\text{edges}} = 494$, shortest tail), consistent with their dispersed, infiltrating spatial distribution; Myeloid cells (Class 3) have the densest ($n_{\text{edges}} = 980$, longest tail).

Hub genes mix cell-intrinsic and neighbourhood signal

The top-degree hubs of each per-class GGM are biologically interpretable, but a careful reading reveals a more nuanced picture than a clean separation between cell-intrinsic and neighbourhood-derived signal. In several classes the canonical markers of the cell type are present in the top-15

hubs alongside markers of spatially adjacent cell types; in other classes the cell-intrinsic markers are largely diluted out and the neighbourhood signal dominates. I describe each class in turn, with reference to the verified top-15 hubs ranked by degree, and to canonical marker expression patterns separately validated in the expression data.

B cells (Class 0) hub on DES, ACTG2, MYH11, COL1A2, SMTN, MYLK, CDC20B, SYNM, ACTA2, PPP1R14A, CNN1 — a smooth-muscle cell signature. The canonical B-cell marker MS4A1 (CD20) does appear at degree 12, but the dominant hubs are an SMC programme. B cells in tumour tissue frequently localise within tertiary lymphoid structures, which are perivascular and surrounded by smooth-muscle vasculature; the GGM picks up the strong SMC programme of this neighbourhood with the B-cell intrinsic signal compressed to a single survivor (MS4A1) in the hub set.

Endothelial cells (Class 1) are a low-abundance population whose canonical markers (PECAM1, VWF, KDR, CDH5) do not appear in the top-15 hubs at all. The dominant hubs are a mix of inflammatory (CXCL8, S100A9, S100A8, IL1B, G0S2, MMP1, MMP3, MMP12, IL1RN) and vascular SMC (MYH11, DES, MYLK, SYNM) signatures — the two cell populations that surround endothelial cells in vascularised tumour tissue. The endothelial GGM has the lowest maximum degree of all eight classes (max degree 20 vs. degree 38 in B cells), consistent with both the low cell count and the heterogeneous neighbourhood.

Fibroblasts (Class 2) hub on CXCL8, COL1A2, S100A9, CXCL5, G0S2, S100A8, PLEK, IL1B, PTGS2, MT2A, MMP1, CCL3 — a mix of canonical fibroblast/CAF markers (COL1A2 at degree 31, the second highest in the network) and an inflammatory/neutrophil signature (CXCL8 at degree 32, S100A8/A9, IL1B, MMP1, CCL3, G0S2). This is one of the cleaner cases of cell-intrinsic and neighbourhood signal coexisting: COL1A2 is a fibroblast marker, the inflammatory genes likely reflect adjacency to inflammatory myeloid cells in inflamed regions of the tumour microenvironment.

Myeloid cells (Class 3) hub on G0S2, CXCL8, S100A9, CSF3R, S100A8, SOCS3, BCL2A1, FOS, COL1A2, DUSP1, TREM1, PTGS2, IL1B, CXCL5, SELL — an activated myeloid/granulocyte signature that matches their cell-intrinsic identity directly. CSF3R, BCL2A1, FOS, SOCS3, TREM1, and SELL are all canonical myeloid markers; the inflammatory chemokines (CXCL8, CXCL5) are also produced by myeloid cells themselves. This is the cleanest case where the cell-intrinsic and neighbourhood signal converge rather than compete: myeloid cells cluster spatially with other myeloid populations, so diffusion reinforces the intrinsic programme.

Normal epithelium (Class 4) has the most heterogeneous hub set: COL1A2 (at degree 21), SLC26A3 (degree 13), MFAP4, DEFA6, IL7R, PTGS2, IGFBP5, G0S2, ADM, LTB, SRGN, ACTA2, FXYD6, TNXB, CXCL8. SLC26A3 and FXYD6 are canonical intestinal-epithelial markers; DEFA6 is a Paneth-cell marker; COL1A2, MFAP4, and TNXB are stromal markers. The hubs reflect both the cell-intrinsic epithelial programme (with intestinal differentiation markers surviving) and the lamina-propria fibroblast adjacency on which intestinal epithelium sits. This is the only class where matrix proteins and canonical epithelial markers appear together in the top-15.

SMC (Class 5) hubs on S100A9, DES, G0S2, CXCL8, MYH11, S100A8, ACTG2, SELL, EGR1, FOSB, DUSP1, FOS, CXCR2, MT2A, FCAR. The canonical SMC markers DES, MYH11, and ACTG2 appear at top positions (degrees 27, 24, 23 respectively), confirming the cell-intrinsic SMC programme. The strong co-occurrence with inflammatory hubs (CXCL8, S100A8/A9, FOS, FOSB, EGR1, CXCR2) reflects the perivascular position of SMC, which places them at the interface with extravasating myeloid cells. SMC is therefore the converse of B cells (Class 0): in B cells the diffusion was from SMC into B cell, here the diffusion is from inflammatory cells into SMC, but the neighbourhood relationship is the same.

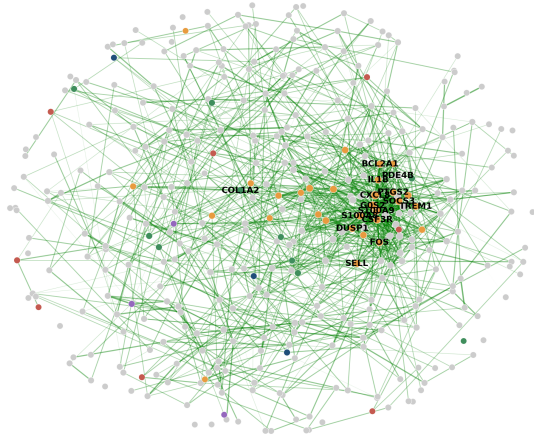
T cells (Class 6) have the most distinctive gene network of all eight classes. Their top hubs are CTSG, MMP25, F13A1, DEFA6, KLHL4, AOC3, IGHM, RNF150, SFRP2, CHRDL2, PNCK, CHIT1, CLCA1, IGLC1, REG3A — a mixed mast-cell/granulocyte/plasma-cell/Paneth-cell signature. No canonical T-cell marker (CD3D, CD3E, CD8A, CD4, TRAC) appears in the top-15. T cells in tumour tissue are infiltrating populations: they appear scattered in low density throughout multiple compartments, with no consistent spatial neighbourhood. The "mixed" hub signature is a consequence of this spatial heterogeneity — the GGM averages over many distinct microenvironments and the surviving hubs are those that appear across multiple unusual neighbourhoods, not the common signal of any single one. T cells are also one of the classes for which the Graphical Lasso did not converge in 200 iterations, consistent with heterogeneity-driven instability. Notably the maximum degree in the T-cell network is only 10 (vs. 38 in B cells, 41 in Myeloid cells), the lowest of all classes, reflecting the lack of a consistent dominant signal.

Tumor (Class 7) hubs on CXCL8, COL1A2, IGFBP5, TIMP3, ELN, SFRP4, IL1B, S100A9, MGP, MT2A, SRGN, ACTG2, SCG2, SIGLEC5, CCL5 — a CAF-rich extracellular matrix signature (COL1A2, TIMP3, ELN, MGP, SFRP4) mixed with inflammatory mediators (CXCL8, IL1B, S100A9, CCL5). Canonical tumour-cell markers (EPCAM, KRT8) are present in the expression data but do not dominate the GGM hubs. Tumor cells in colorectal cancer are typically embedded in CAF-rich stroma with active inflammatory infiltration; both signals coexist in the conditional-dependence network.

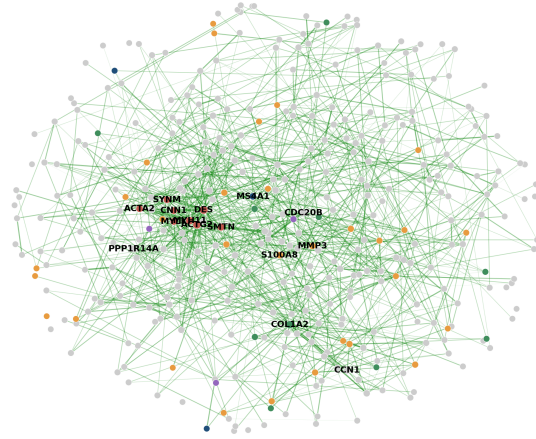
A few hubs appear across multiple classes, suggesting genes that are broadly expressed throughout the tumour microenvironment: CXCL8 appears as a hub in 7 of 8 classes, COL1A2 in 6 of 8, S100A8 and S100A9 in 6 of 8, G0S2 in 5 of 8, MYH11 in 4 of 8. These are inflammatory and matrix-remodelling signatures that contaminate many spatial neighbourhoods through diffusion or are genuinely expressed by multiple cell populations; the GGM cannot distinguish the two without an independent measure of cell-intrinsic transcription.

Figure 16 visualises the contrast between two representative classes: Myeloid cells (Class 3) and B cells (Class 0). The top row shows each full GGM on all 378 common genes; edge thickness is log-scaled by the magnitude of the inverse covariance entry, so the strongest conditional dependencies emerge as visible threads through the cloud of weaker edges. The bottom row reduces each network to its top 35 hubs and lays them out in a circular order sorted by biological family. The asymmetry between the two classes is striking: Myeloid hubs concentrate in an inflammatory arc (orange) where the cell-intrinsic and neighbourhood signals reinforce each other, while B-cell hubs are dominated by SMC markers (red) with MS4A1, the only canonical B-cell hub, sitting in isolation as the lone blue node next to the SMC cluster. The figure is the cleanest visual rendition of the diffusion-mediated reading of Part 4.

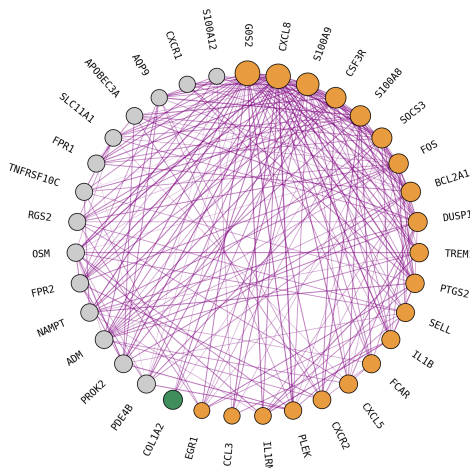
Myeloid cells (c3): full GGM (378 nodes, log-weighted edges)



B cells (c0): full GGM (378 nodes, log-weighted edges)



Myeloid cells (c3): top 35 hubs



B cells (c0): top 35 hubs

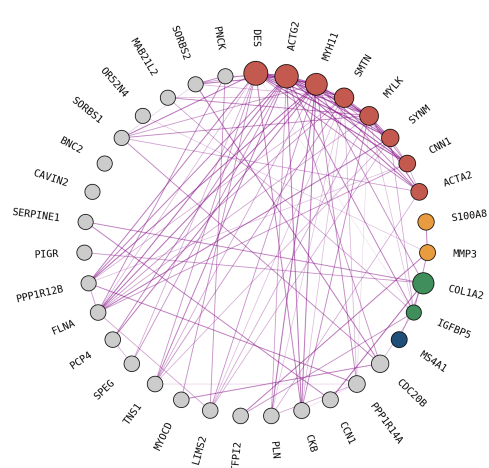


Figure 16: Gene networks of two contrasting cell-type classes: Myeloid cells (Class 3, left) and B cells (Class 0, right). **Top row:** full GGM on the 378 common genes, drawn as a force-directed network. Edge thickness and opacity are log-scaled by $|\Theta_{ij}|$, so the strongest conditional dependencies emerge from the cloud of weaker edges. Nodes are coloured by gene family (see legend); the top 15 hubs by degree are labelled. **Bottom row:** the same network reduced to its top 35 hubs by degree, displayed as a chord diagram sorted by family. The Myeloid network is dominated by inflammatory hubs (orange) that reinforce both cell-intrinsic identity and neighbourhood signal; the B-cell network is dominated by SMC markers (red), with MS4A1 (the only canonical B-cell hub) appearing as a single isolated blue node, consistent with diffusion of mRNA from spatially adjacent smooth muscle cells in perivascular tertiary-lymphoid microenvironments.

Networks share neighbourhoods but rewire connections

The question now is whether the eight networks resemble each other. I use two pairwise similarity measures, computed on the off-diagonals of 8×8 matrices: the Jaccard overlap of the top-15 hubs by degree (*hub Jaccard*) and the Jaccard overlap of the entire edge set (*edge Jaccard*). Figure 17 shows both.

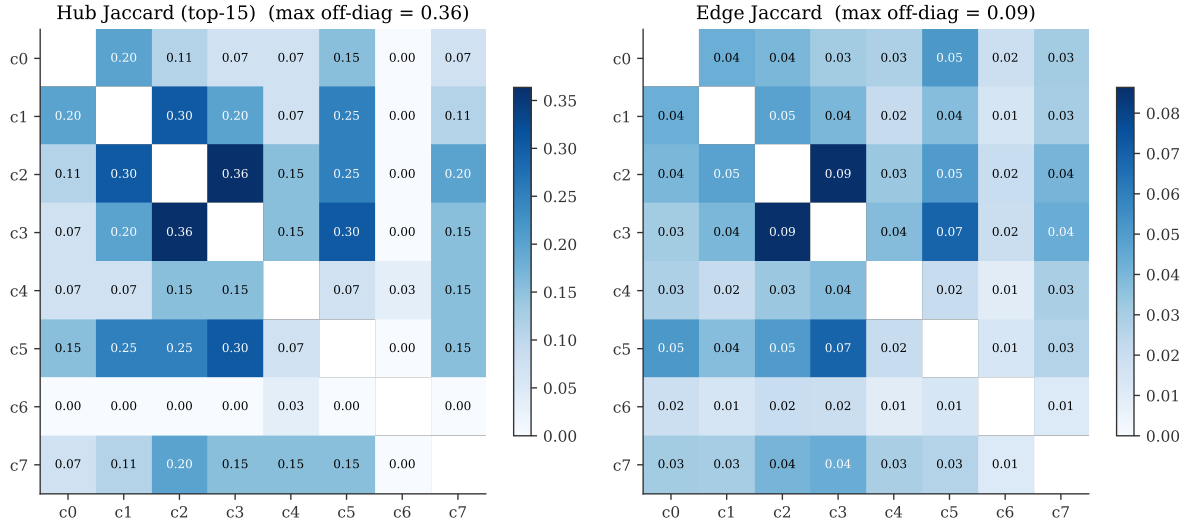


Figure 17: Pairwise hub and edge Jaccard between the 8 per-class GGMs. Left: hub Jaccard on top-15 hubs (max off-diagonal 0.36). Right: edge Jaccard on the full edge sets (max off-diagonal 0.09). Diagonals are masked. The hub-level overlap is up to four times larger than the edge-level overlap, indicating that cell types share microenvironmental hubs but not the conditional-dependence relationships between them. T cells (Class 6) are a structural outlier, with Jaccard ≤ 0.03 to every other class.

Two observations dominate. First, the maximum hub Jaccard among any pair of classes is 0.36 (Fibroblasts vs. Myeloid cells), and most pairs lie in the range 0.07–0.30. Cell types share microenvironmental hubs in moderate but non-trivial proportions. The cluster of B cells (Class 0), Fibroblasts (Class 2), Myeloid cells (Class 3), and SMC (Class 5) is clearly visible as a block of mutual hub overlap: these are the cell types that share the inflammatory/stromal microenvironment of the tumour stroma. Tumor (Class 7) and Normal epithelium (Class 4) sit somewhat apart from this block; T cells (Class 6) are essentially disconnected from all of them.

Second, the maximum edge Jaccard among any pair is only 0.09. The same Fibroblast–Myeloid pair that shares 36% of hubs shares only 9% of edges. Across the full set of pairs, edge Jaccard is roughly an order of magnitude smaller than hub Jaccard. The conclusion is the cleanest finding of Part 4:

Key finding

Cell types share hub genes that reflect spatial neighbourhoods (hub overlap up to 36%) but rewire the specific co-expression relationships between them (edge overlap up to 9%). In some classes (Myeloid, Fibroblasts, SMC, Tumor, Normal epithelium) the cell-intrinsic markers survive as hubs alongside the neighbourhood signal; in others (B cells, Endothelial, T cells) the neighbourhood dominates. The class-specific signal lives in the edges, not in the nodes.

This is biologically sensible. Genes like CXCL8 and S100A8/A9 are broadly distributed inflammatory mediators across the tumour microenvironment, and they appear as hubs in

any class that sits near an inflammatory neighbourhood. The specific partners they connect to, however, depend on which other genes are co-active in that microenvironment. A B-cell's "inflammatory hub" is conditioned on the SMC-rich vascular context of tertiary lymphoid structures; a myeloid cell's "inflammatory hub" is conditioned on other myeloid neighbours. The hubs are shared; the edges differ. The "rewired connections" finding is the genuine class-specific signal; the "shared regulators" finding is, in this dataset, a partial artefact of diffusion combined with a small set of broadly active inflammatory and matrix-remodelling programmes.

T cells are a microenvironmental outlier

T cells are the most striking outlier in Figure 17: their row and column are essentially 0 on both panels. Hub Jaccard between T cells and the other 7 classes is at most 0.03 (one shared hub out of 15 with Normal epithelium, the gene DEFA6); edge Jaccard is at most 0.02. Whatever microenvironmental structure T cells carry, it does not overlap with any of the other classes.

Under the diffusion-contamination framing, this is the consequence of T cells' *lack of a consistent spatial neighbourhood*. T cells in tumour tissue are infiltrating and dispersed: they appear in low density spread across every compartment of the tissue, with no consistent partner cell type. The result is that their microenvironmental gene programme averages over many distinct local contexts, and the dominant signal that survives is the one shared across all the unusual neighbourhoods that T cells happen to find themselves in — not the common signature of any one neighbourhood. The convergence failure of the Graphical Lasso on this class is consistent with this picture: heterogeneous neighbourhoods produce unstable covariance estimates.

This re-interpretation of the T-cell outlier — microenvironmental dispersion rather than cell-intrinsic distinctness — aligns with the topological role identified in Part 1 (T cells sit at the boundary between dense and sparse regions, with negative degree assortativity) and the niche role identified in Part 2 (T cells never dominate a niche but appear as a secondary minority in Niche 17 only). All three signatures point to the same underlying biology: T cells are a spatially diffuse population, and any per-class analysis that depends on consistent spatial context will flag them as distinctive.

Bootstrap stability

To confirm that the headline findings are not an artefact of the particular 5000-cell subsample, I refit each class network on 10 bootstrap resamples (sampled with replacement from the training pool) and tracked hub identity, edge counts, and edge stability. The canonical hub genes (DES, MYH11, ACTA2 for B cells; CXCL8, S100A8/9 for Fibroblasts; COL1A2, TIMP3 for Tumor) have coefficient of variation below 0.20 across the 10 bootstraps. Edge counts reproduce within $\pm 3\text{--}5\%$ of the headline value. Edge identity is less stable: only 22%–34% of the headline edges appear in at least 8 of 10 bootstraps. The conclusion is that the microenvironmental hubs are stable, edge counts are stable, but individual edges have non-trivial variance — as expected from a sparse-precision-matrix estimator at moderate sample size.

Summary. Per-class GGMs at $\alpha = 0.07$ on a 378-gene common subset recover gene networks whose top-degree hubs mix cell-intrinsic markers and diffusion-mediated signal from spatially adjacent cell types. Some classes (Myeloid, Fibroblasts, SMC, Tumor, Normal epithelium) retain canonical cell-type markers among their hubs; others (B cells, Endothelial, T cells) are dominated by neighbourhood signal. Across the 8 networks, hub Jaccard reaches 0.36 (Fibroblasts vs. Myeloid cells) while edge Jaccard stays below 0.09: classes share hubs but rewire connections. T cells are a structural outlier (hub and edge Jaccard ≤ 0.03 to all others), consistent with their dispersed infiltrating spatial distribution and the absence of a consistent neighbourhood.

6 Per-niche gene networks

Part 4 fit gene networks per cell-type class. Part 5 fits them per niche. The two analyses operate on the same data (the cohort’s expression matrix), the same gene subset (378 common genes), the same regulariser ($\alpha = 0.07$), and the same subsampling protocol (5000 cells, patient-stratified). The only thing that changes is the partition: cells are now grouped by the niche assignment from Part 2, not by cell-type label. The question is whether the 20 resulting networks track the composition-based niche families identified in Part 2, or whether the gene-network space carries information that diverges from compositional similarity.

A second motivation is methodological. Niches are heterogeneous in composition (each is a mixture of cell types, with one dominant class and several minorities), and they vary widely in how broadly they recur across patients. If the niche-level networks reproduce the per-class structure of Part 4, that is reassurance: gene networks are robust to the partition used. If they diverge, the divergence itself is informative.

Twenty niche networks at $\alpha = 0.07$

For each of the 20 niches I subsample 5000 cells from the training pool (or fewer if the niche has fewer cells available), restrict to the 378 common genes from Part 4, and fit the Graphical Lasso. The training pool is the same 40-patient training set used everywhere else; the held-out test patients are not used in any GGM. Niche 15 (pure SMC, rare) is available in only 7 of the 40 training patients; I note this as a caveat but proceed with its fit.

A practical issue showed up during fitting: for some niches, several of the 378 common genes are effectively silent within that niche (per-gene standard deviation below 10^{-6} , despite being broadly variable across the full cohort). The Graphical Lasso fails numerically if any input column has zero variance. I therefore drop silent genes per niche before fitting, then re-expand the resulting sparse precision matrix to the original 378×378 shape with an identity diagonal on the excluded entries. This preserves the index alignment across niches and allows pairwise comparison of edge sets.

The number of silent genes per niche varies from 5 to 32, and the variation is itself biologically informative. Niches with restrictive cell-type compositions have more silent genes: Niche 5 (88% Normal epithelium, restricted to 22 patients) silences 32 of the 378 common genes, the most of any niche; Niche 15 (87% SMC, restricted to only 11 patients) silences 24 genes, the second most. Niches with broad, heterogeneous compositions have few silent genes: Niche 17 (the B-cell/T-cell mediator, 48% B cells + 29% T cells) silences only 5 genes. The edge counts of the fitted networks range from 629 to 1087, but the more interpretable quantity is edges-per-kept-gene, which ranges from 1.77 to 3.00. After accounting for the silent-gene effect, the niche networks have comparable connection density per included gene; only their absolute size varies.

Niche networks resemble each other much less than class networks did

The pairwise edge Jaccard between niche networks (Figure 18, right panel) is markedly lower than for the per-class networks. The maximum off-diagonal edge Jaccard is 0.12 — only slightly above the per-class maximum of 0.09 — but the typical pair value is 0.02–0.05. Hub Jaccard reaches up to 0.58, larger than the per-class maximum of 0.36, and composition cosine reaches 1.00 on near-identical-composition niches. The three heatmaps span three different scales of similarity, and reading them at the same colour intensity would be misleading; I rescale each to its own off-diagonal maximum to preserve the internal structure.

The composition heatmap (right panel) shows the strong block structure expected from Part 2: niches with the same dominant class are tightly grouped (the upper-left block contains Tumor-

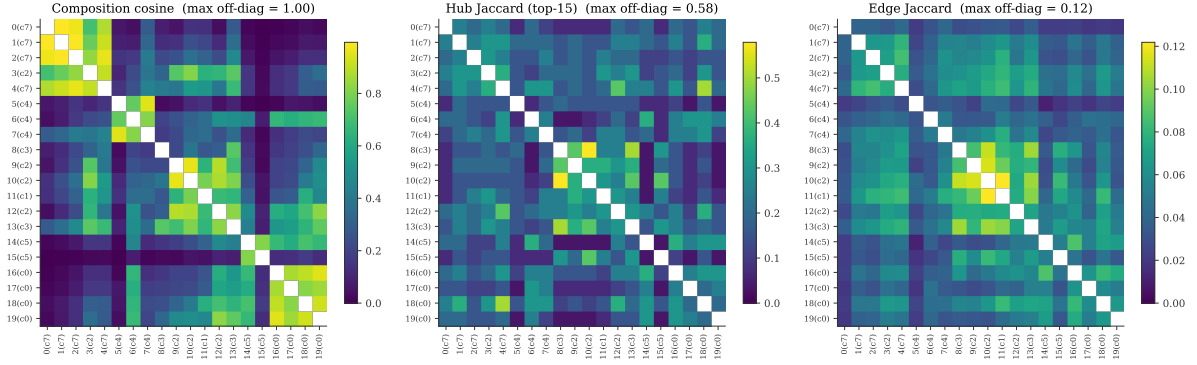


Figure 18: Pairwise similarity between the 20 per-niche networks under three measures: composition cosine (left, max 1.00), top-15 hub Jaccard (middle, max 0.58), and edge Jaccard (right, max off-diagonal 0.12). Diagonals are masked. Niches labelled by id and dominant class. The composition heatmap shows strong block structure (c7-niches in the upper-left, c0-niches in the lower-right). The hub heatmap shows weaker but visible block structure. The edge heatmap shows almost no block structure: niches share regulators much more than they share specific edges.

dominant niches, the lower-right block contains B-cell-dominant niches, and so on). The hub heatmap (middle) shows a weaker but visible echo of this block structure: niches Niche 8–Niche 13, the Myeloid-Fibroblast-Endothelial inflammatory cluster, share more hubs with each other than with the Tumor- or B-cell-dominant blocks. The edge heatmap (right) shows almost no block structure — the bright spots in the middle (Niche 9–Niche 13) are the only weak indication that compositionally similar niches share any edges at all.

Pairwise concordance vs. cluster recovery: a tension

The above suggests a moderate but real correspondence between gene networks and compositional similarity. Quantifying it: across all 190 niche pairs, the Spearman correlation between edge Jaccard and composition cosine is $\rho = 0.615$. The same correlation between hub Jaccard and composition cosine is $\rho = 0.400$. These are non-trivial: niches with similar compositions tend to have more similar gene networks.

However, this is a pairwise statement. When I run hierarchical clustering on each similarity matrix and compare the resulting partition (at $K = 8$, matching the 8 cell-class-dominant families from Part 2) against the Part 2 families using the Adjusted Rand Index (ARI), the picture changes:

- Composition cosine clustering vs. Part 2 families: ARI = 0.802 (sanity check; high as expected)
- Edge Jaccard clustering vs. Part 2 families: ARI = 0.250
- Hub Jaccard clustering vs. Part 2 families: ARI = 0.054

Pairwise concordance does not translate into cluster recovery. The gene networks contain enough information that pairs of similar niches tend to be flagged correctly ($\rho = 0.615$), but not enough to reproduce the family structure of Part 2 as a partition. The dendrograms in Figure 19 make this visually clear: the composition dendrogram (right) is shallow and cleanly partitioned into composition families; the hub dendrogram (middle) has intermediate height and weak clustering; the edge dendrogram (left) is essentially flat, with leaf-to-leaf distances all in the range 0.85–0.97.

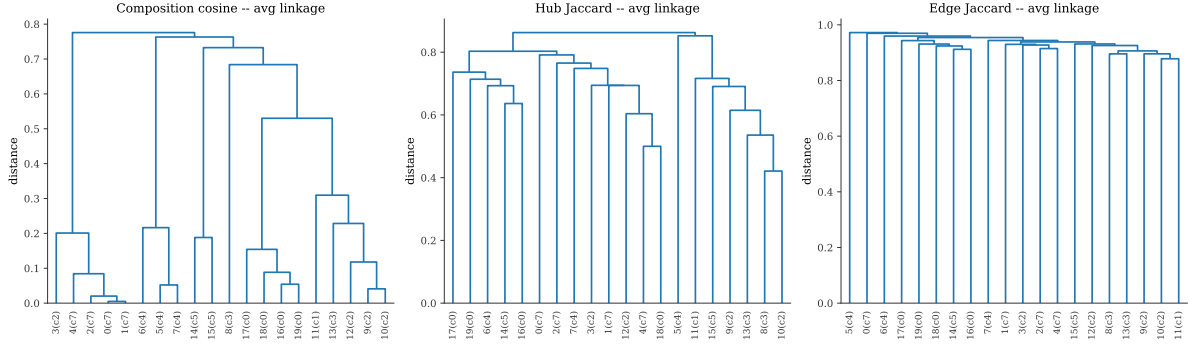


Figure 19: Hierarchical clustering of the 20 niches under three similarity measures, with average linkage. The composition dendrogram (left) is shallow and cleanly grouped: niches with the same dominant class merge at low distance (< 0.30). The hub dendrogram (middle) has intermediate height (0.4–0.85) and only weak family-level structure. The edge dendrogram (right) is flat (heights 0.85–0.97): in edge space, almost all niches are equally distant from each other.

Key finding

Gene networks share some signal with composition (Spearman $\rho = 0.615$ at the pair level) but do not recover the compositional family structure at the cluster level (ARI = 0.25 for edges, 0.05 for hubs, vs. 0.80 for composition itself). Two niches that share a dominant class do tend to have more similar gene networks — but the specific edges connecting genes are largely niche-specific.

This is the "shared regulators, rewired connections" finding from Part 4, replayed at the niche level. At the class level, the pattern was clean: hub Jaccard up to 0.36, edge Jaccard up to 0.09. At the niche level it is weaker but consistent: pairs of compositionally similar niches share some hubs, but the connection structure between hubs varies across niches even when their dominant cell type is the same.

Patient overlap is not the underlying driver

A natural concern is that patient cohort overlap is the latent variable behind both gene-network similarity and compositional similarity. Two niches present in the same set of patients might appear similar in both measures simply because they share the same expression-level batch effects.

To test this, I compute the patient-set Jaccard between each pair of niches (the Jaccard overlap of the set of patients containing at least one cell of that niche, see Figure 20). I then correlate this patient-Jaccard with the other measures:

- Patient Jaccard vs. composition cosine: $\rho = 0.575$
- Patient Jaccard vs. edge Jaccard: $\rho = 0.419$
- Patient Jaccard vs. hub Jaccard: $\rho = 0.367$

Patient overlap correlates with composition (0.575) almost as strongly as with the edge networks (0.419). But the comparison that matters is whether the network–composition correlation (0.615) is stronger than the network–patient correlation (0.419). It is. After accounting for the fact that compositionally similar niches tend to occur in overlapping patient sets, the residual network–composition signal remains the dominant one. The gene-network structure recovers biology more than it recovers cohort overlap.

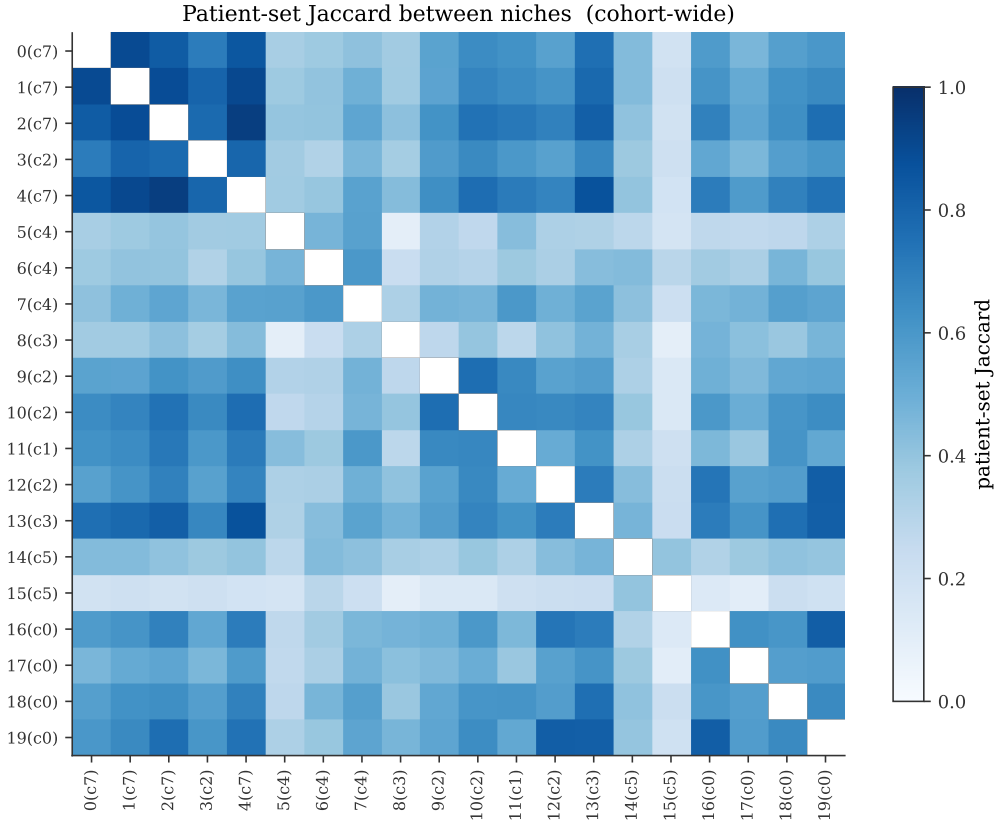


Figure 20: Patient-set Jaccard between the 20 niches (cohort-wide). Niches Niche 0–Niche 4 form a high-overlap block (Jaccard > 0.7); four of the five are Tumor-dominant, and the fifth (Niche 3) is the Fibroblast-Tumor border niche. The rare niches Niche 15 and Niche 14 have low overlap with everything else. The B-cell-dominant niches Niche 16–Niche 19 form a second high-overlap block. Patient overlap correlates with composition ($\rho = 0.575$) and with edge Jaccard ($\rho = 0.419$), but the composition–edge correlation ($\rho = 0.615$) is the stronger of the two: the gene-network structure tracks composition more closely than patient cohort effects.

Niche 17 is a network outlier consistent with its TLS-like composition

Looking at the edge-Jaccard clustering more carefully reveals one striking exception. Niche 17, the B-cell-dominant niche with substantial T-cell secondary presence (48% B cells, 29% T cells, 7% Endothelial cells), splits off into its own singleton in the edge-clustering tree (Figure 19, right). The other B-cell-dominant niches (Niche 16, Niche 18, Niche 19) cluster together at moderate distance, but Niche 17 sits apart from them. In hub-Jaccard clustering it also splits off into its own subtree.

This is consistent with what Part 2 and Part 4 suggested. Niche 17 is the only niche in the cohort where T cells appear as a substantial secondary population — a composition consistent with tertiary lymphoid structures, where B and T cells coexist in organised aggregates. Other B-cell-dominant niches (Niche 16 at 79% B cells, Niche 19 at 53%, Niche 18 at 47%) lack the T-cell co-population and instead carry minorities of Endothelial cells, Fibroblasts, or Myeloid cells. The presence of T cells in Niche 17 brings their distinctive microenvironmental gene programme (recall from Part 4: T cells are the most disjoint class in the per-class GGM space) into the niche-level fit, and this is enough to push Niche 17’s gene network into its own region of edge-Jaccard space. The cell-class-agnostic, network-based analysis independently flags Niche 17 as anomalous in a manner that aligns with the compositional analysis of Part 2 and the per-class network analysis of Part 4.

Summary. Per-niche GGMs reveal moderate pairwise concordance with composition ($\rho = 0.615$ edge–composition) but do not recover the compositional family structure at the cluster level (ARI 0.25). Niches share some hubs (hub Jaccard up to 0.58) but rewire most of their edges (edge Jaccard up to 0.12, typical pair 0.02–0.05). Patient cohort overlap explains some of the structure but is not the dominant driver. Niche 17, the B-cell + T-cell niche (likely tertiary lymphoid structure), emerges independently as a network outlier in the edge-Jaccard clustering, consistent with the distinctive microenvironmental gene programme of T cells identified in Part 4. Gene networks at the niche level recapitulate the class-level "shared regulators, rewired connections" pattern of Part 4, with the additional finding that cluster-level structure is more divergent than pair-level similarity would suggest.

7 Discussion

The five Parts of the project pursue different questions through different methods — graph topology, community detection, supervised learning, sparse precision estimation, cross-similarity analysis — but several themes recur across them. This section pulls them together and discusses what they say about the dataset, the methods, and the limits of what can be concluded from the analysis.

The spatial graph is a productive abstraction at every scale

The most consistent finding across the project is that the spatial KNN graph is a productive level of abstraction at every scale of analysis. Part 1 extracts cohort-level structure from it (architectural bimodality, composition–topology correlations, betweenness skeletons). Part 2 uses it as the substrate for Leiden community detection, which delivers 20 stable niches with biologically interpretable composition. Part 3 feeds it to a GraphSAGE network and gains 6.5 percentage points of validation macro- F_1 over a strong per-cell baseline. Parts 5–6 fit gene networks on the niches that the graph induces, recovering biologically meaningful hub programmes.

No analysis required moving away from the KNN-with-radius-cap construction. This is not a trivial observation: in spatial transcriptomics there is a continuous design choice between Delaunay triangulation, KNN, fixed-radius graphs, learned graphs, and various hybrid constructions, each with different topological properties. The simple KNN-with-radius-cap used here proved sufficient for every analysis in the project. The implication is that the binding constraint on what can be learned from this kind of data is not graph construction; it is what the data themselves contain.

T cells are anomalous on every measure — because they are spatially dispersed

If a single result deserves to be flagged as cross-validated by independent analyses, it is the structural distinctness of T cells (Class 6). Each Part identifies it as different in a different way: Part 1 flags it through negative degree assortativity (T cells sit at boundaries between dense and sparse regions); Part 2 observes that T cells never dominate a niche and appear as a substantial secondary minority in only Niche 17, the B-cell-dominant niche likely corresponding to tertiary lymphoid structures; Part 3 reports T cells as the worst-performing class across every rung of the ablation, with peak $F_1 \approx 0.42$ versus ≥ 0.88 for the best class; Part 4 finds that the T-cell gene network shares essentially zero structure with the seven other classes (top-15 hub Jaccard ≤ 0.03 , edge Jaccard ≤ 0.02). The Graphical Lasso also fails to converge on T cells within 200 iterations.

The unifying explanation is biological. T cells in tumour tissue are infiltrating populations: they appear scattered in low density throughout multiple compartments (tumour, stroma, immune-rich regions), with no consistent spatial partner cell type. This spatial heterogeneity propagates into every downstream analysis. In topology terms (Part 1), it places T cells at the boundary between density modes. In niche terms (Part 2), it prevents T cells from accumulating into a compositionally homogeneous niche of their own. In classification terms (Part 3), it makes T cells the hardest class to predict from neighbourhood features, because their neighbourhood is unstable. In gene-network terms (Part 4), it produces a microenvironmental gene programme that averages over many distinct neighbourhoods and lacks a coherent dominant signal — and that is therefore disjoint from every other cell type’s network. The Graphical Lasso convergence failure is the statistical signature of this heterogeneity: the covariance structure of T cells does not stabilise on a single dominant pattern within the iteration budget.

Notably, T cells emerge in only one niche as a substantial secondary population: Niche 17,

where they coexist with B cells at 29% to 48% proportions. This composition is consistent with tertiary lymphoid structures, organised B/T-cell aggregates that form in tumour tissue as part of the anti-tumour immune response. The fact that this is the *only* place where T cells form a non-trivial population component, and that the same niche emerges in Part 5 as a network outlier disjoint from the other B-cell-dominant niches, is a coherent biological story: T cells in this dataset are diffuse infiltrators with one organised exception (the TLS-like niche), and that exception is independently flagged by every cell-class-blind analysis.

Shared hubs, rewired connections at multiple scales

A second cross-cutting finding is that the "shared hubs, rewired connections" pattern of gene networks emerges at both the cell-type level (Part 4) and the niche level (Part 5). At both levels, top-degree hub genes overlap moderately between similar groups (max hub Jaccard 0.36 for classes, 0.58 for niches), while the specific edges between those hubs overlap much less (max edge Jaccard 0.09 for classes, 0.12 for niches). The shared hubs across classes (CXCL8, S100A8/A9, COL1A2 among the universals) are best interpreted, in the diffusion-contamination framing of Part 4, as gene programmes that are broadly distributed across the tumour microenvironment: they appear as hubs in any class whose neighbourhood includes inflammatory cells or extracellular matrix. The "rewired connections" finding is the genuine class-specific signal; the "shared hubs" finding is partly a property of the data (broadly active programmes) and partly an artefact of mRNA diffusion across cell boundaries at near single-cell resolution.

A subsidiary finding from Part 5 sharpens this: *pairwise* network similarity correlates moderately with compositional similarity ($\rho = 0.615$), but *cluster-level* structure recovery is poor (ARI 0.25 for edges, 0.05 for hubs). The networks contain enough signal to flag pairs of similar niches but not enough to reproduce the compositional family partition. This is a useful methodological caution: a Spearman correlation of 0.6 between two distance matrices is not the same as agreement at the partition level, and confusing the two is a common error in cross-modality analyses.

Compact and fragmented architectures impose downstream constraints

The bimodal architecture observation from Part 1 (largest connected component fraction ~ 0.95 vs. ~ 0.35) has practical consequences throughout the project. Niche detection in Part 2 works well on compact samples ($\sim 97\%$ cell coverage) but only partially on fragmented samples ($\sim 55\%$). The GNN of Part 3 is trained on a `train_mask` that is itself density-biased (Spearman $\rho = -0.91$ between mask retention and mean degree). Per-niche GGMs in Part 5 subsample from the training pool, which under-represents fragmented samples.

These constraints are not failures — the analyses are still internally valid — but they bound the cohort-level generalisation. A reader of this report should take away that the headline numbers (test macro- $F_1 = 0.6089$, 20 stable niches, edge Jaccard $\rho = 0.615$ vs. composition) are valid summaries of the cohort as analysed, but they will tend to overrepresent compact-architecture tissue. Fragmented samples are present in the test set, but their internal structure is less precisely characterised by the niche framework.

The graph subsumes the H&E embedding for cell typing

A more specific methodological observation from Part 3 is that the UNI H&E embedding contributes $\sim 8.5\text{pp}$ of validation F_1 inside a multi-modal MLP, but contributes nothing when added on top of GraphSAGE. The two modalities therefore overlap in the information they carry about cell-type identity. This is not surprising biologically: the H&E embedding captures

local tissue architecture (cell density, packing, adjacency to other cells), which is precisely what the spatial graph propagates through neighbour aggregation. Whichever of the two modalities is present first saturates the architectural signal; the second adds little.

For practitioners, this suggests that on cell-typing tasks where a spatial graph is available, expensive foundation-model embeddings of the H&E image may be a poor return on compute. Conversely, in datasets where the spatial graph is unavailable or unreliable, the H&E embedding partially substitutes. The two are alternative entry points to roughly the same architectural information.

Diffusion contamination shapes what the gene networks can tell us

A property of the data that becomes visible only in Parts 5–6 but applies to every per-cell analysis in the project is *mRNA diffusion contamination*: at the near single-cell resolution of Visium HD, mRNA from a target cell and its ~ 1 –2 nearest spatial neighbours is difficult to disambiguate, and the expression vector of a cell is partly a vector of its spatial neighbourhood. This was confirmed externally by inspection of canonical marker gene dotplots, which show the expected diagonal pattern (each cell type expressing its canonical markers) overlaid on substantial off-diagonal signal from neighbouring cell types.

The implication for Part 4 is that the per-class gene networks recover not pure cell-intrinsic regulatory programmes but *microenvironmental gene programmes* — a mix of the cell's own regulation and the gene expression of cells it typically sits next to. The strength of this mixing varies by class: Myeloid cells, Fibroblasts, SMC, Tumor, and Normal epithelium retain some canonical markers among their hubs (their cell-intrinsic signal survives because they cluster spatially with similar populations); B cells, Endothelial cells, and T cells are dominated by neighbourhood signal because they sit in heterogeneous spatial contexts. The biological interpretations developed in Part 4 (B cells inheriting SMC signal from perivascular TLS contexts; Tumor cells inheriting CAF signal from the surrounding stroma; T cells unable to commit to any consistent neighbourhood) are consistent with this picture and with the spatial biology of tumour microenvironments.

This is not a methodological flaw of the project but a property of the platform. Spatial transcriptomics at near single-cell resolution will always carry some neighbourhood contamination, and the right way to read the per-class and per-niche GGMs is as conditional-dependence networks of *neighbourhoods*, not of *cell types in isolation*. The "shared hubs, rewired connections" finding is robust to this reading: pairs of cell types that share spatial neighbourhoods share gene network hubs, but the specific conditional-dependence edges remain class-specific. A future iteration of this analysis on the same platform would benefit from deconvolution methods (Bayesian or matrix-factorisation-based) that attempt to separate cell-intrinsic from neighbourhood signal at the read level; in the current report, the diffusion is acknowledged and incorporated into the interpretation rather than corrected.

What I cannot claim

A few qualifications are worth stating explicitly.

First, cell-type hub interpretations. The interpretations of hub gene programmes in Part 4 are inferences about what the GGM is recovering — a mixture of cell-intrinsic regulation and diffusion-mediated neighbourhood signal — and should be read as biologically plausible but not validated. In particular, the interpretation of Niche 17 as a tertiary lymphoid structure follows from its composition (B cells dominant, T cells substantial secondary, low frequency in the cohort) but is not confirmed by independent immunohistochemistry or transcript-level markers of TLS organisation.

Second, the niche count $K = 20$. This was chosen by inspection of the dendrogram and intra-cluster homogeneity. A nearby choice ($K = 15$ or $K = 25$) would merge or split some of the smaller niches but is unlikely to change the qualitative findings. I did not perform a formal stability analysis on K .

Third, the Graphical Lasso convergence failures. Endothelial cells, Fibroblasts, and T cells did not reach the dual-gap tolerance within 200 iterations for the per-class fits, and several per-niche fits had the same issue. The results from non-converged iterates appear stable in bootstrap, but I cannot rule out that a more careful regularisation schedule (or a different sparse-precision estimator) would yield different edge sets. The qualitative findings (hub identity, hub vs. edge Jaccard, class–niche concordance) are robust across the bootstrap resamples, which is the strongest stability evidence I have.

Fourth, the test-validation gap. Validation macro- $F_1 = 0.6350$ vs. test macro- $F_1 = 0.6089$ (−2.6pp gap) is small but non-zero. With only 8 test patients, the per-patient variance is non-trivial ($\sigma = 0.042$, range $[0.473, 0.625]$). A larger test set would tighten the headline estimate, but the current configuration is the standard one for the dataset.

Fifth, the Gaussianity assumption of the Graphical Lasso. Log-normalised spatial transcriptomics counts are zero-inflated and non-Gaussian; restricting to broadly variable common genes mitigates the issue at the marginal level but does not eliminate it. The principled alternative is the nonparanormal SKEPTIC estimator [10], which uses rank correlations and is consistent under arbitrary continuous marginal transformations. A SKEPTIC refit would be the natural next step for any extension that derives quantitative conclusions from the precision-matrix entries; the qualitative findings of Part 4–Part 5 (hub identity, shared-hubs vs. rewired-connections, T-cell outlier) are likely robust but not formally validated against the more defensible estimator.

Directions the analysis points toward

A few questions are left open. The poor per-niche performance on niche-pure regions (Niche 0, Niche 1, Niche 15 all below median test F_1) suggests that a niche-conditioned cell-typing approach could help: instead of one classifier for all niches, train a niche-specialised head per family. Part 5 already provides the per-niche partition needed for this. The differential gene-network analysis between paired similar niches (e.g. Niche 17 vs. Niche 16/Niche 18/Niche 19, all B-cell-dominant) is also a natural extension: the differential edges between two compositionally similar niches localise the niche-specific regulatory rewiring that Part 5 only captures at the aggregate level. A complementary direction is to compare the gene network of the B-cell-only Niche 16 against the B-cell-plus-T-cell Niche 17: the network differences would isolate the conditional-dependence signature of the T-cell co-population, with potential implications for TLS organisation in tumour tissue. Both directions are well-motivated by the findings of the current analysis and would build directly on the artefacts already produced.

Closing remark. The project demonstrates that a single dataset of ~ 8 million spatially-resolved cells supports a wide range of network-theoretic analyses: graph topology, community detection, neighbourhood-aware classification, and conditional-dependence networks of genes. The most informative result is not any individual headline number; it is the consistency with which the same patterns (T cells as the spatially dispersed outlier, shared-hubs rewired-connections, architectural bimodality, the dominance of the graph over per-cell modalities) reappear across independent analyses. The spatial graph is the integrating object, the conditional-independence networks are the most refined level at which biological structure becomes visible, and the diffusion contamination characteristic of high-resolution spatial transcriptomics shapes those networks into descriptions of *microenvironments* rather than of cell types in isolation.

8 Methods

This section collects the technical details deferred from the narrative Parts, organised by stage of the pipeline. The goal is to make the work reproducible from the cached artefacts and source modules without requiring the reader to re-derive design decisions from the codebase.

Data and access

The dataset is provided as a Zarr v3 store. Each cell carries: a 2D spatial position, a sample identifier (0–111), a patient identifier (0–55), a cell-type label (0–7), a 1000-dimensional HVG expression vector (selected per the procedure below), a 1280-dimensional UNI [2] H&E embedding, four morphology scalar features, and a binary `train_mask` flag. The total cell count is 8,065,561 across 112 samples; 66,796 cells are isolated (no spatial neighbours within the cap radius) and are dropped from spatial-graph-based analysis. All Zarr access uses the `oindex` API; cells are loaded in chunks of $\sim 10^5$ at a time to bound peak memory.

The canonical patient split is fixed once and persisted to `cache/splits/patient_split.json`. It is generated with random seed 42 from the full 56-patient pool: 40 training patients, 8 validation, 8 test. Test patients are $\{0, 21, 26, 28, 34, 40, 45, 51\}$. This split is loaded at the beginning of every notebook and every training script, ensuring that no analysis is computed on test patients except for the final evaluation in §4.

Spatial graph construction

For each sample I compute a symmetric k -nearest-neighbour graph with $k = 8$ in 2D Euclidean space, then cap edges at a per-sample radius threshold computed as $1.5\times$ the 99th-percentile nearest-neighbour distance in that sample. The cap removes edges that bridge tissue gaps in fragmented samples; it has no effect on dense compact samples. The resulting graph is symmetrised: an edge (i, j) is kept if j is among i 's k nearest within the cap, or i is among j 's. The implementation uses scikit-learn's `BallTree` for the KNN query, then constructs the edge index with NumPy operations. Per-sample subgraphs are cached as `.npz` files under `cache/per_sample/`.

Topology metrics

The free-tier metrics are computed with `python-igraph` on the per-sample subgraph. They include: mean degree, fraction of isolated cells, fraction of cells in the largest connected component, number of connected components of size ≥ 100 , global transitivity, mean local clustering coefficient, degree assortativity, and cell-type assortativity. Anomaly detection flags a sample as anomalous if mean degree < 4 AND fraction of isolated nodes > 0.35 AND largest-component fraction < 0.1 . Two samples (#87, #107) match all three criteria.

The expensive tier (closeness and betweenness centrality) is computed on the giant connected component using `igraph`'s `betweenness` with source/target sampling [3]. I use 5,000 source nodes uniformly sampled from the giant component (no cutoff distance), which produces stable centrality rankings on a graph of size $\sim 40,000$ at a fraction of the cost of full computation. Closeness is similarly computed on the giant component with full source-target enumeration (this is cheaper than betweenness on the same graph). Ten representative samples form the expensive-tier subset; the per-node arrays are not persisted by default and are recomputed when needed for figures.

Niche detection

Per-sample Leiden [4] is run with `leidenalg` on the weighted giant-component subgraph. Edge weights are $\exp(-\|x_i - x_j\|^2/2\sigma^2)$ where x_i, x_j are the HVG expression vectors and σ is the median pairwise distance among the edge endpoints in that sample. The resolution parameter is $\gamma = 0.5$ for the Reichardt–Bornholdt partition modularity, selected from a sweep over $\{0.3, 0.5, 0.7, 1.0\}$ on three representative samples (#2, #47, #83) using intra-cluster composition homogeneity as the criterion. Random seed for each Leiden run is the sample index.

Cross-sample alignment is performed by hierarchical clustering on the per-community composition vectors. I retain communities with at least 100 cells (9,457 of 11,496 non-anomaly communities) to ensure that the composition vector is statistically reliable. The clustering uses Ward linkage [11] with Euclidean distance on the 8-dimensional composition vectors. The number of niches $K = 20$ is chosen as the smallest K at which all clusters have median intra-cluster cosine distance ≤ 0.15 , ensuring compositional purity. Niche labels are assigned by mapping each community to its cluster and propagated to all cells in the community. Cells in communities below the 100-cell threshold remain unassigned (niche label -1).

HVG selection and modality standardisation

The 1000 highly variable genes used in Parts 4–6 are selected with the Seurat criterion as implemented in `scanpy` [6] (function `highly_variable_genes` with `flavor='seurat'`). Selection is performed on training cells only, on log-normalised expression. The output index is cached at `cache/preprocessing/hvg_indices.npy`.

Per-modality standardisation uses training-cell statistics only. HVG expression is z-scored gene-wise in a chunked pass to avoid loading the full $1.4 \times 10^6 \times 1000$ training-cell matrix at once. H&E embeddings are similarly z-scored dimension-wise. Morphology features are z-scored on the four scalar dimensions. All standardisation parameters (mean, std) are persisted to `cache/preprocessing/` for reproducible inference on validation and test sets.

GNN training

The GNN implementations are in PyTorch Geometric [12]. GraphSAGE uses two convolution layers of hidden dimension 256 with mean aggregation; GAT uses two layers, 4 attention heads, 64 hidden dimensions per head. Both have dropout 0.2 on intermediate activations. Modality-specific encoders (used in Rungs 2, 4, 5) project each modality to 256 dimensions before concatenation; the head is a single linear layer to 8 logits.

Training is performed with `NeighborLoader` from PyG, using [20,20] neighbours per layer (capped 2-hop sampling). Batch size is 2048 during training and 4096 during evaluation. Optimisation uses Adam with learning rate 10^{-3} , weight decay 10^{-4} , and 20 maximum epochs with early stopping (patience 3) on validation macro- F_1 . Training is run on a single NVIDIA A40 (24GB) and takes 5–30 minutes per rung depending on architecture. Per-rung pickles are saved to `cache/models/` with full history, best state, and final-epoch validation and test predictions.

For Rungs 1 (logistic regression) and 2 (multi-modal MLP), I use the same training/validation/test split and HVG selection but skip the spatial-graph data loading. Rung 1 uses `scikit-learn`’s `LogisticRegression` with L_2 penalty and `lbfgs` solver, fit on the full training set in one pass. Rung 2 is an MLP with the same per-modality encoders as the GNN rungs but no graph convolution; it is trained with the same Adam setup as the GNN rungs.

The modality-masking analysis on Rung 2 zeroes out a modality’s output at inference time and

recomputes validation F_1 on the entire validation set. The reported ΔF_1 values are differences from the unmasked Rung 2 baseline.

Graphical Lasso fitting

A property of high-resolution spatial transcriptomics relevant to Parts 5–6 is mRNA diffusion contamination: at Visium HD resolution, transcripts assigned to a target cell partially originate from cells within ~ 1 –2 neighbours of it. I do not apply a deconvolution correction to the expression vectors before fitting the Graphical Lasso; instead, I read the resulting conditional-dependence networks as *microenvironmental gene programmes* that mix cell-intrinsic regulation with neighbourhood signal, and incorporate this framing into the biological interpretation throughout Parts 5–6 and the Discussion. The decision to fit the GGM on uncorrected expression is deliberate — the diffusion-aware reading of the networks is informative in itself — and the validation of cell-type identities against canonical marker-gene expression patterns confirms that the canonical markers are present in each class, even where the GGM hubs are dominated by neighbourhood signal.

A second known limitation of the Graphical Lasso on count-based data is the multivariate Gaussian assumption, which spatial-transcriptomics expression vectors do not satisfy (zero-inflation, right-skew, mean-dependent variance). The common-gene restriction (per-gene std > 0.05 in every cell-type class) selects genes with sufficiently rich variability across cells that the marginal distributions, after log-normalisation, are less degenerate than the typical scRNA-seq gene; this is the practical mitigation. The principled alternative is the nonparanormal SKEPTIC estimator [10], which replaces sample correlations with rank-based correlations (Kendall’s τ or Spearman’s ρ) and is consistent under arbitrary continuous marginal transformations. I do not apply it in this report and acknowledge it as the more defensible estimator for any extension that derives quantitative conclusions from the precision-matrix entries.

For Parts 5–6 the Graphical Lasso is fit with scikit-learn’s `GraphicalLasso` (with `max_iter=200` and default tolerance 10^{-4}). The input to each fit is the log-normalised expression of 5,000 subsampled cells, restricted to the 378 common genes. The common-gene subset is the genes among the 1000 HVGs whose per-gene standard deviation exceeds 0.05 in *every* cell-type class.

Per-class fits in Part 4 use patient-stratified sampling: cells are drawn proportionally from each training patient that contributes at least 50 cells of that class. Per-niche fits in Part 5 use the same protocol on the niche assignment. Niche 15 has training data from only 7 patients, which I flag as a caveat. Random seed for subsampling is fixed per niche/class (the niche/class index) for reproducibility.

Within each fit, genes with standard deviation below 10^{-6} on the subsample are dropped before passing to the estimator (these are genes that are broadly variable across the cohort but happen to be silent in this particular niche or class subsample, e.g. class-specific markers in classes that don’t express them). After fitting, the resulting precision matrix is re-expanded to the original 378×378 shape with identity diagonal on the excluded rows/columns, so all networks share a common gene-space index for pairwise comparison.

The $\alpha = 0.07$ choice is the result of a sweep over $\{0.05, 0.07, 0.097, 0.136, 0.189, 0.264, 0.368, 0.514, 0.717, 1.0\}$ on SMC (Class 5) and Tumor cells (Class 7), chosen as two biologically distant classes (mesenchymal/vascular vs. epithelial/tumour) for the calibration. The criterion was edge sparsity near 1% of $\binom{378}{2}$ combined with biological hub interpretability. The sweep was performed once and the resulting α is reused across all subsequent fits.

Bootstrap stability for GGMs

The bootstrap stability analysis in Part 4 re-fits each per-class GGM on 10 bootstrap resamples of the original 5000-cell subsample, drawn with replacement. For each bootstrap, I record the top-15 hub gene identities and the edge set at $\alpha = 0.07$. Stability of a hub gene is the coefficient of variation of its degree across bootstraps (lower is more stable); stability of an edge is the fraction of bootstraps in which it appears. The bootstrap pickles are stored at `cache/ggm/bootstraps_alpha_007_common378.pkl`.

Cross-niche similarity measures

In Part 5, the three pairwise similarity matrices over the 20 niches are computed as follows. Edge Jaccard is $|E_a \cap E_b|/|E_a \cup E_b|$ where E_a, E_b are the off-diagonal non-zero entries of the precision matrices of niches a and b . Hub Jaccard is the analogous formula on the top-15 hub gene sets (top by degree). Composition cosine is the cosine similarity of the 8-dimensional cell-type composition vectors of the two niches. Patient-set Jaccard is the Jaccard overlap of the sets of patients containing at least one cell of that niche.

Hierarchical clustering on each matrix uses average linkage on the distance matrix $1 - S$ (where S is the similarity matrix); for composition cosine the distance is $1 - \cos$. The dendrograms are cut at $K = 8$ to compare against the 8 cell-class-dominant family partition from Part 2, using the Adjusted Rand Index [13].

Computing environment and caching

All computations were run on a single workstation with an NVIDIA A40 GPU (24GB), 134GB of system RAM, and a 250GB home filesystem (with overflow to a 466GB local scratch directory). Python 3.12, PyTorch 2.12 with CUDA 13, PyTorch Geometric with `torch_sparse` and `torch_scatter` (no `pyg-lib` due to wheel unavailability for this Torch/CUDA combination). Per-niche GNN cohort caches (~ 100 GB total) are symlinked from the home directory to local scratch.

Caching is hierarchical: `cache/global/` holds the cohort-wide arrays (cell positions, classes, samples, patients, train mask) re-stored as memory-mapped NumPy files; `cache/per_sample/sample_NNN.npz` holds per-sample spatial subgraphs; `cache/topology/`, `cache/communities/`, `cache/preprocessing/`, `cache/models/`, `cache/ggm/`, `cache/ggm_niches/` hold the artefacts of each Part. Most cells in the analysis notebooks load from cache rather than recomputing; the only routine that runs end-to-end on every report build is the figure generation script in `scripts/generate_report_figures.py`.

9 References

- [1] 10x Genomics. *Visium HD Spatial Gene Expression*. <https://www.10xgenomics.com/products/visium-hd-spatial-gene-expression>. 2024.
- [2] Richard J. Chen et al. “Towards a general-purpose foundation model for computational pathology (UNI)”. In: *Nature Medicine*. 2024.
- [3] Ulrik Brandes and Christian Pich. “Centrality estimation in large networks”. In: *International Journal of Bifurcation and Chaos* 17.7 (2007), pp. 2303–2318.
- [4] V. A. Traag, L. Waltman, and N. J. van Eck. “From Louvain to Leiden: guaranteeing well-connected communities”. In: *Scientific Reports* 9.1 (2019), p. 5233. DOI: [10.1038/s41598-019-41695-z](https://doi.org/10.1038/s41598-019-41695-z).
- [5] Tim Stuart, Andrew Butler, Paul Hoffman, et al. “Comprehensive integration of single-cell data”. In: *Cell* 177.7 (2019), pp. 1888–1902.
- [6] F. Alexander Wolf, Philipp Angerer, and Fabian J. Theis. “SCANPY: large-scale single-cell gene expression data analysis”. In: *Genome Biology* 19.1 (2018), p. 15.
- [7] William L. Hamilton, Rex Ying, and Jure Leskovec. “Inductive Representation Learning on Large Graphs”. In: *Advances in Neural Information Processing Systems* 30 (2017).
- [8] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. “Graph Attention Networks”. In: *International Conference on Learning Representations* (2018).
- [9] Jerome Friedman, Trevor Hastie, and Robert Tibshirani. “Sparse inverse covariance estimation with the graphical lasso”. In: *Biostatistics* 9.3 (2008), pp. 432–441. DOI: [10.1093/biostatistics/kxm045](https://doi.org/10.1093/biostatistics/kxm045).
- [10] Han Liu, Fang Han, Ming Yuan, John Lafferty, and Larry Wasserman. “High-dimensional semiparametric Gaussian copula graphical models”. In: *The Annals of Statistics* 40.4 (2012), pp. 2293–2326.
- [11] Fionn Murtagh and Pierre Legendre. “Ward’s Hierarchical Agglomerative Clustering Method”. In: *Journal of Classification* 31.3 (2014), pp. 274–295.
- [12] Matthias Fey and Jan Eric Lenssen. “Fast Graph Representation Learning with PyTorch Geometric”. In: *ICLR Workshop on Representation Learning on Graphs and Manifolds*. 2019.
- [13] Lawrence Hubert and Phipps Arabie. “Comparing partitions”. In: *Journal of Classification* 2.1 (1985), pp. 193–218.